Signals and Systems

Complete Lecture Notes & Core Formulations

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I. Signals

1. Classification of Signals

Amplitude \ Time	Continuous Time	Discrete Time
Continuous Amplitude	Analog Signal	Sampled Signal
Discrete Amplitude	Quantized Signal	Digital Signal

2. Signal Decomposition

① Convergence Properties (Spectral & Basis Expansions in Dirac Bra-Ket Notation):

1. **Continuous-Time, Continuous-Spectrum Decomposition:**

$$f(t) = \int d\lambda \, |a_\lambda\rangle\langle a_\lambda|f(t)\rangle$$

Examples: Time domain ($t/\delta(t)$); Laplace Transform (LT/e^{st}); Fourier Transform ($\mathrm{FT}/e^{j\omega t}$).

2. **Continuous-Time, Discrete-Spectrum (Periodic) Decomposition:**

$$f(t) = \sum_i |a_i\rangle\langle a_i|f(t+mT)\rangle$$

Examples: Time domain ($t/\delta(t)$); Fourier Series ($\mathrm{FS}/e^{jn(\frac{2\pi}{T})t}$); LT, FT.

3. **Discrete-Time, Continuous-Spectrum Decomposition:**

$$f(n) = \int d\lambda \, |a_\lambda\rangle\langle a_\lambda|f(n)\rangle$$

Examples: Discrete time ($n/\delta(n)$); z-Transform (ZT/z^{n-1}); Discrete-Time Fourier Transform ($\mathrm{DTFT}/e^{jn\omega}$).

4. **Discrete-Time, Discrete-Spectrum (Periodic) Decomposition:**

$$f(n) = \sum_i |a_i\rangle\langle a_i|f(n+rN)\rangle$$

Examples: Discrete time ($n/\delta(n)$); Discrete Fourier Series ($\mathrm{DFS}/e^{j(\frac{2\pi}{N})nk}$); $\mathrm{ZT}, \mathrm{DTFT}$.

Orthogonality and Completeness (Parseval's Identity):

- **Orthogonality:** $\langle a_\lambda|a_{\lambda'}\rangle = \delta(\lambda-\lambda'), \quad \langle a_i|a_j\rangle = \delta_{ij}$
- **Completeness:**

$$\int dt \, \langle f(t)|f(t)\rangle = \int d\lambda \, \langle a_\lambda|a_\lambda\rangle, \quad \int dt \, \langle f(t)|f(t)\rangle = \sum_i \langle a_i|a_i\rangle$$

$$\sum_n \langle f(n)|f(n)\rangle = \int d\lambda \, \langle a_\lambda|a_\lambda\rangle, \quad \sum_n \langle f(n)|f(n)\rangle = \sum_i \langle a_i|a_i\rangle$$

② Correlation Properties:

- **Energy Signals:**
  1. Continuous-time: $R_{12}(\tau) = \langle f_2(t-\tau)|f_1(t)\rangle = R_{21}^*(-\tau)$, with energy $E_1 = R_{11}(0)$.
  2. Discrete-time: $R_{12}(m) = \langle f_2(n-m)|f_1(n)\rangle = R_{21}^*(-m)$, with energy $E_1 = R_{11}(0)$.
- **Power Signals:**
  3. Continuous-time: $R_{12}(\tau) = \lim_{T\to\infty} \frac{1}{T}\langle f_2(t-\tau)|f_1(t)\rangle_T = R_{21}^*(-\tau)$, with average power $P_1 = R_{11}(0)$.
  4. Discrete-time: $R_{12}(m) = \lim_{N\to\infty} \frac{1}{N}\langle f_2(n-m)|f_1(n)\rangle_N = R_{21}^*(-m)$, with average power $P_1 = R_{11}(0)$.

II. Systems

1. Classification of Systems

Property	Class I	Class II
Parameter Distribution	Lumped: ODE / Difference Eq.	Distributed: PDE / Partial Diff. Eq.
Time Domain	Continuous: Differential Eq.	Discrete: Difference Eq.
Memory	Memoryless: Algebraic Eq.	With Memory: Differential / Difference Eq.
Linearity	Linear: Superposition & Homogeneity	Nonlinear: Non-superposition or Non-homogeneous
Time-Invariance	Time-Varying: $x(t-t_0) \nrightarrow y(t-t_0)$	Time-Invariant: $x(t-t_0) \to y(t-t_0)$
Invertibility	Invertible: Injective & Surjective (Bijective)	Non-invertible: Non-injective or Non-surjective
Causality	Causal: $x(t-t_0) \to y(t)$ $(t_0 \geq 0)$	Non-causal: $x(t+t_0) \to y(t)$ $(t_0 > 0)$
Stability	Stable: BIBO	Unstable: $\overline{\text{BIBO}}$

Primary Focus: Linear Time-Invariant (LTI) systems with deterministic inputs, lumped parameters, linearity, and time-invariance.

- **Models:** Input-output external models vs. state-variable internal models.
- **Transform Domains:**
  - Continuous: $t \to s \to \Omega \to \Omega_0$ (Time → Laplace → Continuous Fourier → Fourier Series)
  - Discrete: $n \to z \to \omega \to \omega_0$ (Discrete Time → z-Transform → DTFT → DFS/DFT)

2. System Decomposition & Representation

- ① Block diagrams
- ② Signal flow graphs and Mason's Gain Formula

Appendix A1: Generalized Functions (Distributions) and Impulse Properties

1. The Dirac Delta Distribution $\delta(t)$

Defined as a functional acting on a test function $\varphi(t)$:

$$N_\delta[\varphi(t)] = \langle \delta(t), \varphi(t) \rangle = \varphi(0)$$

- **Distribution (Generalized Function) $\delta(t)$:** Assigns the value $\varphi(0)$ to $\varphi(t)$.
- **Test Function $\varphi(t)$:** Smooth function of compact support $\varphi(t) \in C^\infty(\mathbb{R})$ (i.e., $D(\mathbb{R})$), where $\varphi(t) \neq 0$ only on a bounded open interval $a < t < b$ $(-\infty < a, b < \infty)$.

Fundamental Properties of $\delta(t)$:

1. **Linearity:** $a\delta(t) + b\delta(t) = (a+b)\delta(t)$
2. **Sampling (Multiplication) Property:** $f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0)$
3. **Scaling Property:** $\delta(at) = \frac{1}{|a|}\delta(t) \implies \delta(t)$ is an even function ($\delta(-t) = \delta(t)$).
4. **Composition with a Smooth Function $f(t)$:** For simple roots t_i where $f(t_i) = 0$ and $f'(t_i) \neq 0$:

$$\delta[f(t)] = \sum_i \frac{1}{|f'(t_i)|}\delta(t - t_i)$$

2. Higher-Order Derivatives of the Impulse $\delta^{(k)}(t)$

$$N_{\delta^{(k)}}[\varphi(t)] = \langle \delta^{(k)}(t), \varphi(t) \rangle = (-1)^k \varphi^{(k)}(0)$$

This identity is proved via integration by parts and the compact support of $\varphi(t)$. The strength of $\delta(t)$ corresponds to the area enclosed during the limiting approximation process.

- ① **Multiplication by a smooth function $f(t)$:** Can be evaluated to arbitrary order k via integration by parts:

$$\langle f(t)\delta^{(k)}(t), \varphi(t) \rangle = \langle \delta^{(k)}(t), f(t)\varphi(t) \rangle = (-1)^k \frac{d^k}{dt^k}[f(t)\varphi(t)]\Big|_{t=0}$$

- ② **Scaling of $\delta^{(k)}(t)$:**

$$\delta^{(k)}(at) = \frac{1}{|a|}\frac{1}{a^k}\delta^{(k)}(t) \implies \delta^{(k)}(t) \text{ is even for even } k \text{ and odd for odd } k.$$

3. Riemann–Lebesgue Lemma in Distribution Sense

$$N_{e^{-j\omega t}}[\varphi(t)] = \lim_{\omega \to \infty} \int_a^b e^{-j\omega t}\varphi(t)\, dt = 0 \quad (-\infty < a, b < \infty)$$

- ① As $\omega \to \infty$, in the distributional (weak) sense, $e^{-j\omega t} = e^{j\omega t} = \cos(\omega t) = \sin(\omega t) = 0$.

III. Continuous-Time Signals and Systems (t-Domain)

1. Signal Representation via Impulses (Sifting Property)

- ① $f(\tau) = \int_{-\infty}^{\infty} f(t)\delta(\tau - t)\,dt$
- ② $f(t) = \int_{-\infty}^{\infty} f(\tau)\delta(t - \tau)\,d\tau$

2. System Response Solution

① Natural (Free) Response + Forced Response (Classical ODE Solution):

- Given a linear constant-coefficient ordinary differential equation (ODE):
  - **Homogeneous Equation ($e(t) = 0$):** Solve the characteristic equation to obtain the homogeneous solution (natural/free response): $r_h(t) = \sum_k A_k e^{\alpha_k t}$.
  - **Non-homogeneous Equation ($t > 0_+$):** Use the method of undetermined coefficients corresponding to the input form to obtain a particular solution (forced response): $B(t)$.
- Summing the two gives the total response containing undetermined constants A_k: $r(t) = \sum_k A_k e^{\alpha_k t} + B(t)$.
- Using the initial states $x(0_-)$ and the **impulse matching method** on the ODE at $t = 0$, determine the post-impulse initial conditions $x(0_+)$, which then uniquely determine the coefficients A_k for the complete response.

② Zero-State Response (r_{zs}) + Zero-Input Response (r_{zi}):

- **Zero-State Response ($e(t) \to H[e(t)]$):**

$$r_{zs}(t) = \sum_k A_{zsk} e^{\alpha_k t} + B(t)$$

  Subject to initial conditions:

$$\begin{cases} x_{zs}(0_-) = 0 \\ x_{zs}(0_+) \text{ determined via the impulse matching method} \end{cases}$$

  Note: $x_{zs}(t)$ contains $\delta(t)$ (and its derivatives) when the highest derivative order of $e(t)$ (or δ order in $e(t)$) $\geq$ the order of the differential equation; determined via impulse matching.

- **Zero-Input Response ($x(0_+) \to H[x(0_+)]$):**

$$r_{zi}(t) = \sum_k A_{zik} e^{\beta_k t}$$

  Subject to initial conditions:

$$\begin{cases} x_{zi}(0_-) \text{ is given} \\ x_{zi}(0_+) = x_{zi}(0_-) \quad (\text{since } x_{zi}(t) \text{ contains no } \delta) \end{cases}$$

- **Total Response Decomposition & Linearity:**

$$
\begin{aligned}
r(t) &= [r_{zs}(t) + r_{zi}(t)]u(t) = H[e(t)] + H[x(0_+)] \\
&= H\left[\int_{-\infty}^{\infty} e(\tau)\delta(t-\tau)\,d\tau\right] + H[x(0_+)] \\
&= \int_{-\infty}^{\infty} e(\tau)h(t-\tau)\,d\tau + H[x(0_+)] \\
&= e(t) * h(t) + H[x(0_+)]
\end{aligned}
$$

Exhibits **zero-state linearity** (with respect to excitation $e(t)$) and **zero-input linearity** (with respect to initial state $x(0_+)$).

3. System Impulse Response $h(t)$ Properties

- ① **Invertibility:** $h(t) * \tilde{h}(t) = \delta(t)$
- ② **Causality:** $h(t) = h(t)u(t)$ (i.e., $h(t) = 0$ for $t < 0$)
- ③ **Stability (BIBO):** Suppose $|e(t)| \le B < \infty$:

$$
|r(t)| = \left|\int_{-\infty}^{\infty} h(\tau)e(t-\tau)\,d\tau\right| \le \int_{-\infty}^{\infty} |h(\tau)|\,|e(t-\tau)|\,d\tau \le B\int_{-\infty}^{\infty} |h(\tau)|\,d\tau
$$

Therefore, a necessary and sufficient condition for BIBO stability is absolute integrability:

$$
\int_{-\infty}^{\infty} |h(\tau)|\,d\tau < \infty
$$

IV. Laplace Domain Signals and Systems (s-Domain)

1. Signal Representation (Bilateral Laplace Transform)

- ① **Analysis Equation (Direct Transform):**

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}\,dt, \quad s = \sigma + j\omega$$

Region of Convergence (ROC): $\mathrm{ROC} = \left\{ s \;\middle|\; \int_{-\infty}^{\infty} |f(t)|e^{-\sigma t}\,dt < \infty \right\}$, which forms a vertical strip in the s-plane:

$$\begin{cases} \text{Left-sided signal:} & \mathrm{ROC} = \{s \mid -\infty < \sigma < \sigma_r\} \\ \text{Right-sided signal:} & \mathrm{ROC} = \{s \mid \sigma_l < \sigma < \infty\} \\ \text{Two-sided signal:} & \mathrm{ROC} = \{s \mid \sigma_l < \sigma < \sigma_r\} \end{cases}$$

- ② **Synthesis Equation (Inverse Transform):**

$$f(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} F(s)e^{st}\,ds = \sum_{p_i} \mathrm{Res}[F(s)e^{st}, p_i] \quad (\sigma \in \mathrm{ROC})$$

For the same algebraic expression $F(s)$, the time-domain signal depends on the ROC:

$$\begin{cases} \text{If ROC} = \{s \mid -\infty < \sigma < \sigma_r\}, & f(t) = f(t)u(-t) \quad \text{(left-sided)} \\ \text{If ROC} = \{s \mid \sigma_l < \sigma < \infty\}, & f(t) = f(t)u(t) \quad \text{(right-sided)} \\ \text{If ROC} = \{s \mid \sigma_l < \sigma < \sigma_r\}, & f(t) = f_r(t)u(t) + f_l(t)u(-t) \quad \text{(two-sided)} \end{cases}$$

2. System Response & Partial Fraction Expansion

Under zero-state conditions, $r(t) = e(t) * h(t)$, so in the s-domain:

$$R(s) = E(s) \cdot H(s), \quad \mathrm{ROC}(R) \supseteq \mathrm{ROC}(H) \cap \mathrm{ROC}(E)$$

For a system described by a linear constant-coefficient ODE, $H(s)$ is a rational function. By the Fundamental Theorem of Algebra:

$$H(s) = \frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{j=1}^{n}(s - p_j)}$$

- ① **Distinct Real Poles:**

$$\begin{cases} m < n \text{ (Proper):} & \text{Partial fraction expansion with residues } K_i = (s - p_i)H(s)|_{s=p_i} \\ m \geq n \text{ (Improper):} & \text{Long division first; note } \mathcal{L}^{-1}(s^k) = \delta^{(k)}(t) \end{cases}$$

- ② **Complex Conjugate Poles:** Can be expanded either in real quadratic form or complex conjugate first-order terms:

$$\begin{cases} \dfrac{1}{s^2 + p_1^2} \cdot \dfrac{1}{s + p_2} = \dfrac{as + b}{s^2 + p_1^2} + \dfrac{k}{s + p_2} \\ \dfrac{1}{s^2 + p_1^2} \cdot \dfrac{1}{s^2 + p_2^2} = \dfrac{k_1}{s + jp_1} + \dfrac{k_1^*}{s - jp_1} + \dfrac{k_2}{s + jp_2} + \dfrac{k_2^*}{s - jp_2} \end{cases}$$

- ③ **Multiple (Repeated) Poles of Order k at $s = p_1$:**

$$H(s) = \frac{K_{11}}{(s - p_1)^k} + \frac{K_{12}}{(s - p_1)^{k-1}} + \cdots + \frac{K_{1k}}{s - p_1} + \frac{E(s)}{P(s)}$$

where the coefficients are computed recursively (or via derivatives):

$$K_{11} = H(s)(s-p_1)^k\Big|_{s=p_1} = H_1(s)|_{s=p_1}$$

$$K_{12} = \frac{H_1(s)-K_{11}}{s-p_1}\bigg|_{s=p_1} = H_2(s)|_{s=p_1}, \quad K_{13} = \frac{H_2(s)-K_{12}}{s-p_1}\bigg|_{s=p_1} = H_3(s)|_{s=p_1}, \dots$$

Inverse transform pair: $\mathcal{L}^{-1}\left(\frac{1}{s^n}\right) = \pm\frac{t^{n-1}}{(n-1)!}u(\pm t)$.

3. System Function $H(s)$ Properties

- ① **Invertibility:** $H(s)\cdot\tilde{H}(s) = 1$
- ② **Causality:** For a rational $H(s)$, the ROC is a right half-plane ($\mathrm{Re}(s) > \sigma_{\max}$).
- ③ **Stability:**
  1. **Stable $\Longleftrightarrow$ ROC contains the $j\omega$-axis ($\mathrm{Re}(s)=0$):**

$$\int_{-\infty}^{\infty}|h(t)|\,dt < \infty \Longleftrightarrow \int_{-\infty}^{\infty}|h(t)e^{-st}|\,dt\bigg|_{s=j\omega} < \infty \implies |H(s)|\Big|_{s=j\omega} < \infty$$

  2. **Marginally Stable** $\Longleftrightarrow$ ROC has the $j\omega$-axis as its boundary (with only simple poles on the $j\omega$-axis).
  3. **Unstable** $\Longleftrightarrow$ ROC does not contain the $j\omega$-axis.

Appendix A3: Bilateral Laplace Transform Properties

Property	Time Domain $f(t)$	s-Domain $F(s)$	ROC
① Differentiation / Integration	$\frac{d}{dt} \to s, \quad \int_{-\infty}^{t} \to \frac{1}{s}$	$\frac{d}{ds} \to -t, \quad \int_{s}^{\infty} \to \frac{1}{t}$	Contains R (or $R \cap \{\sigma > 0\}$)
② Time Shifting (t-move)	$f(t - t_0)$	$e^{-st_0}F(s)$	R
③ Frequency Shifting (s-move)	$e^{s_0 t}f(t)$	$F(s - s_0)$	R shifted by $\mathrm{Re}(s_0)$
④ Scaling	$f(at)$	$\frac{1}{\lvert a \rvert}F\left(\frac{s}{a}\right)$	R/a
⑤ Conjugation	$f^*(t)$	$F^*(s^*)$	R
⑥ Convolution	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$	At least $R_1 \cap R_2$
⑦ Initial Value Theorem	If $f(t) = f(t)u(t)$ and $f(t)$ contains $k\delta(t)$, then $f(0_+) = \lim_{s\to\infty}[sF(s) - ks]$		
⑧ Final Value Theorem	If $f(t) = f(t)u(t)$ and $F(s)$ is stable or has at most a simple pole at $p = 0$, then $f(\infty) = \lim_{s\to 0} sF(s)$		
⑨ Correlation	$f_1(t) * f_2(-t)$	$F_1(s) \cdot F_2(-s)$	At least $R_1 \cap (-R_2)$

Appendix A4: Response Classification, All-Pass, and Minimum-Phase Systems

- ① **Transient, Steady-State, and Divergent Responses:**
  - **Transient (Decaying) Response:** $R(s)$ stable $\implies$ both $H(s)$ and $E(s)$ are stable.
  - **Steady-State Response:** $R(s)$ marginally stable $\implies$ $H(s)$ or $E(s)$ is marginally stable while the other is stable.
  - **Divergent Response:** $R(s)$ unstable $\implies$ $H(s)$ or $E(s)$ is unstable.
  - *Special Case:* Pole-zero cancellation between $H(s)$ and $E(s)$.
- ② **Natural (Free) vs. Forced Response in s-Domain:**
  - **Natural Response:** Contributed by the poles of $H(s)$.
  - **Forced Response:** Contributed by the poles of $E(s)$.
  - *Special Case:* When $H(s)$ and $E(s)$ share common poles, in the partial fraction expansion of $R(s)$, the terms whose pole orders match the pole orders in $H(s)$ belong to the natural response.
- ③ **All-Pass and Minimum-Phase Systems:**
  - **First-order all-pass:** $H_1(s) = \frac{s - s_p}{s + s_p^*}$; **Second-order all-pass:** $H_2(s) = \frac{s - s_p}{s + s_p^*} \cdot \frac{s - s_p^*}{s + s_p}$; **General all-pass:** $H_{ap}(s) = \prod_i H_{1i}(s) \prod_j H_{2j}(s)$.
  - **All-pass group delay property:** $-\frac{d}{d\omega}\varphi_{ap}(\omega) > 0$.

- **Minimum-Phase System $H_{\min}(s)$:** All poles and zeros lie in the left half-plane (or on the $j\omega$-axis).
- **Canonical Decomposition:** Any stable, causal non-minimum-phase system can be factored as:

$$H(s) = H_{ap}(s) \cdot H_{\min}(s)$$

- **Properties:** Group delay satisfies $-\frac{d}{d\omega}\varphi(\omega) > -\frac{d}{d\omega}\varphi_{\min}(\omega)$ (minimum-phase has minimum group delay). Furthermore, if $|H(j\omega)| = |H_{\min}(j\omega)|$, partial energy satisfies $\int_0^t |h_{\min}(\tau)|^2 d\tau \geq \int_0^t |h(\tau)|^2 d\tau$ (written $\langle h_{\min}(t)|h_{\min}(t)\rangle \geq \langle h(t)|h(t)\rangle$).

Appendix A5: Unilateral Laplace Transform Properties

Definition: $F(s) = \int_{0_-}^{\infty} f(t)e^{-st}\,dt = \int_{-\infty}^{\infty} f(t)u(t)e^{-st}\,dt$, with inverse $f(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st}\,ds$.

- ① **Differentiation & Integration:**

$$\mathcal{L}\left[\frac{d^n}{dt^n}f(t)\right] = s^nF(s) - s^{n-1}f(0_-) - s^{n-2}f'(0_-) - \cdots - s^0 f^{(n-1)}(0_-)$$

$$\mathcal{L}\left[\int_{-\infty}^{t} f(\tau)\,d\tau\right] = \frac{F(s)}{s} + \frac{\int_{-\infty}^{0_-} f(\tau)\,d\tau}{s}$$

- ② **Convolution:** When both $f_1(t)$ and $f_2(t)$ are causal, $f_1(t) * f_2(t) \to F_1(s)F_2(s)$.
- ③ **Time Shifting (t-move):** $f(t-t_0)u(t-t_0) \to F(s)e^{-st_0}$ ($t_0 > 0$).

V. Continuous-Time Fourier Domain (ω-Domain Signals & Systems)

1. Signal Representation (Continuous-Time Fourier Transform)

- ① **Direct Transform:**

$$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt \quad \text{(Laplace transform evaluated on the } j\omega\text{-axis)}$$

Relation to $F(s)$ based on $\mathrm{ROC}(F(s))$:

$$\begin{cases} \mathrm{ROC}(F(s)) \text{ contains } j\omega\text{-axis}: & F(j\omega) = F(s)|_{s=j\omega} \\ \mathrm{ROC}(F(s)) \text{ does not contain } j\omega\text{-axis}: & F(j\omega) \text{ does not exist} \\ \mathrm{ROC}(F(s)) \text{ has } j\omega\text{-axis as boundary}: & \text{If there is a } k\text{-th order pole on the } j\omega\text{-axis at } s = j\omega_0: \end{cases}$$

$$F(s) = F_0(s) + \frac{K_0}{(s-j\omega_0)^k} \implies F(j\omega) = F_0(s)|_{s=j\omega} + \frac{K_0\pi j^{k-1}}{(k-1)!}\delta^{(k-1)}(\omega-\omega_0)$$

- ② **Inverse Transform:**

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\omega)e^{j\omega t}\,d\omega$$

2. System Response & Spectral Density

Under zero-state conditions, $r(t) = e(t) * h(t)$, so:

$$R(j\omega) = E(j\omega)\cdot H(j\omega) \quad \text{if } \mathrm{ROC}(E) \text{ and } \mathrm{ROC}(H) \text{ contain the } j\omega\text{-axis.}$$

- **Energy Signals (Energy Spectral Density $\mathcal{E}(\omega)$ & Autocorrelation):**

$$\mathcal{E}_r(j\omega) = \mathcal{E}_e(j\omega)\cdot|H(j\omega)|^2, \quad R_r(t) = R_e(t) * h(t) * h^*(-t) = R_e(t) * R_h(t)$$

- **Power Signals (Power Spectral Density $P(\omega)$ & Autocorrelation):**

$$P_r(j\omega) = P_e(j\omega)\cdot|H(j\omega)|^2, \quad R_r(t) = R_e(t) * h(t) * h^*(-t) = R_e(t) * R_h(t)$$

3. System Function $H(j\omega)$ Properties

- ① **Invertibility:** $H(j\omega)\cdot\tilde{H}(j\omega) = 1$
- ② **Causality:** For rational $H(j\omega)|_{j\omega=s}$, the ROC is the right half-plane containing the $j\omega$-axis.
- ③ **Stability:** Always BIBO stable (since $H(j\omega)$ exists on the $j\omega$-axis).

VI. Continuous-Time Periodic Fourier Series Domain (ω_0-Domain Signals & Systems)

1. Signal Representation (Fourier Series, $\omega_0 = \frac{2\pi}{T}$)

- ① **Fourier Series Coefficients:**

$$F_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-jn(\frac{2\pi}{T})t}\,dt = \frac{1}{2}\sqrt{a_n^2+b_n^2}\,e^{-j\arctan\frac{b_n}{a_n}}$$

- ② **Fourier Series Expansion:**

$$f(t)=\sum_{n=-\infty}^{\infty}F_ne^{jn(\frac{2\pi}{T})t}=a_0+\sum_{n=1}^{\infty}\left[a_n\cos\left(n\frac{2\pi}{T}t\right)+b_n\sin\left(n\frac{2\pi}{T}t\right)\right]$$

2. System Response via Periodic Extension

Under zero-state conditions, $r(t)=e(t)*h(t)$, so for periodic/extended signals:

$$R_n=E_n\cdot H_n\cdot T\quad\left(\text{or } R_n=E_n\cdot H_n\cdot\frac{1}{T}\text{ depending on normalization}\right)$$

Suppose $e(t)$ has finite duration T_1 and $h(t)$ has finite duration T_2; then $r(t)=e(t)*h(t)$ has duration $T_3=T_1+T_2$. Periodically extending $e(t),h(t),r(t)$ with period $T\geq T_3$ yields $e_T(t),h_T(t),r_T(t)$, whose Fourier Series with fundamental frequency $\omega_0=\frac{2\pi}{T}$ give E_n,H_n,R_n.

3. System Function H_n Properties

- ① **Invertibility:** $H_n\cdot\tilde{H}_n=T$ (or 1), with period $T\geq T_h+T_{\tilde{h}}$.
- ② **Causality:** For rational $H_n|_{j\frac{2\pi}{T}n=s}$, the ROC is a right half-plane containing the $j\omega$-axis.
- ③ **Stability:** Always stable (since $h(t)$ is the periodic extension of a time-limited impulse response h).

Appendix A6: Fourier Transform of Periodic Signals & Wiener–Khinchin Derivation

Let $f(t)$ be periodic with period T ($\omega_0 = \frac{2\pi}{T}$). Then:

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn(\frac{2\pi}{T})t}, \quad F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta\left(\omega - n\frac{2\pi}{T}\right), \quad F_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-jn(\frac{2\pi}{T})t}\,dt = \frac{1}{T}F_T(\omega)|_{\omega=n\frac{2\pi}{T}}$$

Power Spectral Density $P(\omega) = \mathcal{F}[R(\tau)]$ of a Periodic Signal:

$$\begin{aligned}
P(\omega) = \mathcal{F}[R(\tau)] &= \mathcal{F}\left\{\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} f(t)f^*(t-\tau)\,dt\right\} \\
&= \mathcal{F}\left\{\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}\sum_{p=-\infty}^{\infty} F_p e^{jp(\frac{2\pi}{T})t}\sum_{q=-\infty}^{\infty} F_q^* e^{-jq(\frac{2\pi}{T})(t-\tau)}\,dt\right\} \\
&= \mathcal{F}\left\{\sum_{p,q} F_p F_q^* e^{jq(\frac{2\pi}{T})\tau}\left[\lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2} e^{j(p-q)(\frac{2\pi}{T})t}\,dt\right]\right\} \\
&= \mathcal{F}\left\{\sum_{p,q} F_p F_q^* e^{jq(\frac{2\pi}{T})\tau}\delta_{p,q}\right\} = \mathcal{F}\left\{\sum_{q=-\infty}^{\infty}|F_q|^2 e^{jq(\frac{2\pi}{T})\tau}\right\} = 2\pi\sum_{n=-\infty}^{\infty}|F_n|^2\delta\left(\omega - n\frac{2\pi}{T}\right)
\end{aligned}$$

Appendix A7: Continuous-Time Fourier Transform Properties

1. **Duality (Symmetry):**
   $\mathcal{F}[F(t)] = 2\pi f(-\omega) \implies \mathcal{F}^{-1}[F(\omega)]\big|_{t=-\omega} = \frac{1}{2\pi}\mathcal{F}[F(t)] \implies \mathcal{F}^{-1}[f(\omega)] = \frac{1}{2\pi}\mathcal{F}[f(t)]|_{\omega=-t}$
2. **Scaling:** $\mathcal{F}[f(at)] = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)$
3. **Time Shifting (t-move):**
   $\mathcal{F}[f(t-t_0)] = F(\omega)e^{-j\omega t_0} \implies \mathcal{F}[f(at-t_0)] = \frac{1}{|a|}F\left(\frac{\omega}{a}\right)e^{-j\omega t_0/a}$
4. **Frequency Shifting (ω-move):** $\mathcal{F}[f(t)e^{j\omega_0 t}] = F(\omega-\omega_0)$
5. **Differentiation & Integration:**

$$\frac{d^n}{dt^n} \to (j\omega)^n, \quad \frac{d^n}{d\omega^n} \to (-jt)^n, \quad \int_{-\infty}^{t} \to \cdot\,\mathcal{F}[u(t)] = \frac{1}{j\omega} + \pi\delta(\omega), \quad \int_{-\infty}^{\omega} \to \cdot\, 2\pi\mathcal{F}^{-1}[u(\omega)]$$

6. **Convolution & Multiplication:**

$$\mathcal{F}[f_1 * f_2] = F_1 \cdot F_2, \quad \mathcal{F}[f_1 \cdot f_2] = \frac{1}{2\pi}F_1 * F_2$$

$$\mathcal{F}^{-1}[F_1(\omega)\cdot F_2(\omega)] = \mathcal{F}^{-1}[F_1(\omega)] * \mathcal{F}^{-1}[F_2(\omega)], \quad \mathcal{F}^{-1}[F_1 * F_2] = 2\pi\mathcal{F}^{-1}[F_1]\cdot\mathcal{F}^{-1}[F_2]$$

7. **Correlation (Wiener–Khinchin):** $\mathcal{F}[R_{12}(\tau)] = F_1(\omega)\cdot F_2^*(\omega) \implies \mathcal{F}[R(\tau)] = |F(\omega)|^2$

Appendix A8: Reconstructing Continuous Spectrum from Fourier Series

Let $f(t)$ be a time-limited signal of duration T_0 (supported on $[0, T]$). Periodically extending $f(t)$ with period $T \geq T_0$ yields $f_T(t)$:

- FT of $f(t)$: $F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt$
- FS of $f_T(t)$: $F_n = \frac{1}{T}\int_T f_T(t)e^{-jn(\frac{2\pi}{T})t}\,dt$

Since $f(t) = f_T(t)[u(t) - u(t-T)]$, convolving in the frequency domain gives the interpolation formula:

$$F(j\omega) = \frac{1}{2\pi}\left[\sum_{n=-\infty}^{\infty} F_n \cdot 2\pi\delta\left(\omega - n\frac{2\pi}{T}\right)\right] * \left[T \cdot \mathrm{Sa}\left(\frac{\omega T}{2}\right)e^{-j\omega\frac{T}{2}}\right] = T\sum_{n=-\infty}^{\infty} F_n \,\mathrm{Sa}\left[\frac{T}{2}\left(\omega - n\frac{2\pi}{T}\right)\right]e^{-j\frac{T}{2}\left(\omega - n\frac{2\pi}{T}\right)}$$

where $\mathrm{Sa}(x) = \frac{\sin x}{x}$.

VII. Discrete-Time Signals and Systems (n-Domain)

1. Signal Representation via Unit Impulses

- ① $x(n) = \sum_{m=-\infty}^{\infty} x(m)\delta(n-m)$
- ② $x(m) = \sum_{n=-\infty}^{\infty} x(n)\delta(m-n)$

2. Difference Equation Solution

① Natural (Free) Response + Forced Response:

- Given an N-th order linear constant-coefficient difference equation:
  - **Homogeneous Equation ($e(n) = 0$):** Solve the characteristic equation to get the homogeneous solution (natural response): $\sum_k C_k \alpha_k^n$.
  - **Particular Solution ($n \geq 0$):** Use undetermined coefficients matching the input form to obtain the forced response: $D(n)$.
- Summing gives the complete response with undetermined constants C_k: $r(n) = \sum_k C_k \alpha_k^n + D(n)$.
- From pre-initial states $y(-N) \sim y(-1)$, iterate the difference equation to find $y(0) \sim y(N-1)$, which uniquely determine C_k.

② Zero-State Response (r_{zs}) + Zero-Input Response (r_{zi}):

- **Zero-State Response ($e(n) \to H[e(n)]$):**

$$r_{zs}(n) = \sum_k C_{zsk}\alpha_k^n + D(n), \quad \begin{cases} y_{zs}(-N) \sim y_{zs}(-1) = 0 \\ y_{zs}(0) \sim y_{zs}(N-1) \text{ obtained by iteration} \end{cases}$$

- **Zero-Input Response ($y(0) \sim y(N-1) \to H[\cdots]$):**

$$r_{zi}(n) = \sum_k C_{zik}\beta_k^n, \quad \begin{cases} y_{zi}(-N) \sim y_{zi}(-1) \text{ are given} \\ y_{zi}(0) \sim y_{zi}(N-1) \text{ (or solved directly from } y_{zi}(-N) \sim y_{zi}(-1)) \end{cases}$$

  Note: The response to $\delta(n)$ and $\delta(n-n_0)$ in $e(n)$ can be found using LTI time-invariance and the zero-input method (since $h(0) = y(0)$ determines the coefficients).

- **Total Response Decomposition & Linearity:**

$$\begin{aligned} r(n) &= [r_{zs}(n) + r_{zi}(n)]u(n) = H[e(n)] + H[y(0) \sim y(N-1)] \\ &= H\left[\sum_{m=-\infty}^{\infty} e(m)\delta(n-m)\right] + H[y(0) \sim y(N-1)] \\ &= \sum_{m=-\infty}^{\infty} e(m)h(n-m) + H[y(0) \sim y(N-1)] \\ &= e(n) * h(n) + H[y(0) \sim y(N-1)] \end{aligned}$$

  Exhibits both **zero-state linearity** and **zero-input linearity**.

3. Discrete-Time System Impulse Response $h(n)$ Properties

- ① **Invertibility:** $h(n) * \tilde{h}(n) = \delta(n)$

- ② **Causality:** $h(n) = h(n)u(n)$ (i.e., $h(n) = 0$ for $n < 0$)
- ③ **Stability (BIBO):** For bounded input $|e(n)| \leq B < \infty$:

$$|r(n)| = \left| \sum_{m=-\infty}^{\infty} h(m)e(n-m) \right| \leq \sum_{m=-\infty}^{\infty} |h(m)|\,|e(n-m)| \leq B \sum_{m=-\infty}^{\infty} |h(m)|$$

Thus, BIBO stability holds if and only if $h(n)$ is absolutely summable:

$$\sum_{m=-\infty}^{\infty} |h(m)| < \infty$$

Appendix A9: Hilbert Transform Relations for Causal Signals and Sequences

① Causal Continuous-Time Signal ($h(t) = h(t)u(t)$)

Let $H(\omega) = R(\omega) + jX(\omega) = \mathcal{F}[h(t)]$. Since $h(t) = h(t)u(t)$ (assuming no impulse at $t = 0$):

$$\begin{aligned} H(\omega) &= \frac{1}{2\pi}\mathcal{F}[h(t)] * \mathcal{F}[u(t)] = \frac{1}{2\pi}[R(\omega) + jX(\omega)] * \left[\pi\delta(\omega) + \frac{1}{j\omega}\right] \\ &= \frac{1}{2\pi}\left[\pi R(\omega) + X(\omega) * \frac{1}{\omega}\right] + \frac{j}{2\pi}\left[\pi X(\omega) - R(\omega) * \frac{1}{\omega}\right] \\ &= \left[\frac{R(\omega)}{2} + \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{X(\xi)}{\omega - \xi}\,d\xi\right] + j\left[\frac{X(\omega)}{2} - \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{R(\xi)}{\omega - \xi}\,d\xi\right] \end{aligned}$$

Equating real and imaginary parts gives the **Kramers–Kronig / Hilbert Transform relations**:

$$R(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{X(\xi)}{\omega - \xi}\,d\xi, \quad X(\omega) = -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{R(\xi)}{\omega - \xi}\,d\xi$$

② Causal Discrete-Time Sequence ($x(n) = x(n)u(n)$)

A causal sequence can be recovered from its even part $x_e(n)$ or odd part $x_o(n)$:

$$x(n) = 2x_e(n)u(n) - x_e(0)\delta(n) = 2x_o(n)u(n) + x(0)\delta(n)$$

Taking the DTFT, where
$\mathrm{DTFT}[u(n)] = \pi\sum_{k=-\infty}^{\infty}\delta(\omega - 2\pi k) + \frac{1}{1-e^{-j\omega}} = \pi\sum_{k=-\infty}^{\infty}\delta(\omega - 2\pi k) + \frac{1}{2} - \frac{j}{2}\cot\left(\frac{\omega}{2}\right)$:

$$\begin{aligned} X(e^{j\omega}) &= \mathrm{DTFT}[2x_e(n)u(n) - x_e(0)\delta(n)] \\ &= \frac{1}{2\pi}\left[\mathrm{Re}\,X(e^{j\omega}) * 2\pi\sum_{k=-\infty}^{\infty}\delta(\omega - 2\pi k) + \int_{-\pi}^{\pi}\mathrm{Re}\,X(e^{j\omega})\,d\omega - \mathrm{Re}\,X(e^{j\omega}) * j\cot\frac{\omega}{2}\right] - \frac{1}{2\pi}\int_{-\pi}^{\pi}\mathrm{Re}\,X(e^{j\omega})\,d\omega \\ &= \mathrm{Re}\,X(e^{j\omega}) - \frac{j}{2\pi}\mathrm{Re}\,X(e^{j\omega}) * \cot\frac{\omega}{2} \end{aligned}$$

Similarly, from the odd part:

$$X(e^{j\omega}) = \mathrm{DTFT}[2x_o(n)u(n) + x(0)\delta(n)] = \frac{1}{2\pi}\left[j\,\mathrm{Im}\,X(e^{j\omega}) * 2\pi\sum_{k=-\infty}^{\infty}\delta(\omega - 2\pi k) + j\int_{-\pi}^{\pi}\mathrm{Im}\,X(e^{j\omega})\,d\omega - j\,\mathrm{Im}\,X(e^{j\omega}) * j\cot\frac{\omega}{2}\right] +$$

Therefore, the real and imaginary parts of $X(e^{j\omega})$ satisfy the discrete periodic Hilbert relations (where $\mathcal{P}$ denotes the Cauchy principal value):

$$\mathrm{Re}[X(e^{j\omega})] = x(0) + \frac{1}{2\pi}\mathcal{P}\int_{-\pi}^{\pi}\mathrm{Im}[X(e^{j\theta})]\cot\left(\frac{\omega - \theta}{2}\right)d\theta$$

$$\mathrm{Im}[X(e^{j\omega})] = -\frac{1}{2\pi}\mathcal{P}\int_{-\pi}^{\pi}\mathrm{Re}[X(e^{j\theta})]\cot\left(\frac{\omega - \theta}{2}\right)d\theta$$

③ Causal Spectrum (Analytic Signal $f_a(t) = f(t) + j\hat{f}(t)$ with One-Sided Spectrum)

$$f(t) = -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\hat{f}(\tau)}{t - \tau}\,d\tau, \quad \hat{f}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{f(\tau)}{t - \tau}\,d\tau$$

④ Causal Discrete Line Spectrum

Analogous discrete Hilbert relations hold by duality between the time and frequency domains.

VIII. z-Domain Signals and Systems (z-Transform)

1. Signal Representation (Bilateral z-Transform)

- ① **Direct Transform:**

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}, \quad z = re^{j\omega}$$

Region of Convergence (ROC): $\mathrm{ROC} = \left\{ z \mid \sum_{n=-\infty}^{\infty} |x(n) z^{-n}| < \infty \right\}$, forming an annulus (ring) in the z-plane:

$$\begin{cases} \text{Left-sided sequence:} & \mathrm{ROC} = \{ z \mid 0 \le |z| < r_o \} \quad (\text{may exclude } z = 0) \\ \text{Right-sided sequence:} & \mathrm{ROC} = \{ z \mid r_i < |z| \le \infty \} \quad (\text{may exclude } z = \infty) \\ \text{Two-sided sequence:} & \mathrm{ROC} = \{ z \mid r_i < |z| < r_o \} \end{cases}$$

- ② **Inverse z-Transform (Contour Integration & Residue Theorem):**

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} \, dz = \sum_{p_i \text{ inside } C} \mathrm{Res}[X(z) z^{n-1}, p_i] \quad (C \subset \mathrm{ROC})$$

For the same rational $X(z)$, the sequence depends on the ROC:

$$\begin{cases} \text{If ROC} = \{ z \mid 0 \le |z| < r_i \}, & x(n) = x(n) u(-n-1) \quad (\text{left-sided}) \\ \text{If ROC} = \{ z \mid |z| > r_o \}, & x(n) = x(n) u(n) \quad (\text{right-sided}) \\ \text{If ROC} = \{ z \mid r_i < |z| < r_o \}, & x(n) = x_r(n) u(n) + x_l(n) u(-n-1) \quad (\text{two-sided}) \end{cases}$$

2. System Response & Partial Fraction Expansion

Under zero-state conditions, $r(n) = e(n) * h(n)$, so:

$$R(z) = E(z) \cdot H(z), \quad \mathrm{ROC}(R) \supseteq \mathrm{ROC}(H) \cap \mathrm{ROC}(E)$$

For a linear constant-coefficient difference equation, $H(z)$ is rational. Expanding $\frac{H(z)}{z} = \frac{\prod_{i=1}^{m}(z - z_i)}{\prod_{j=1}^{n}(z - p_j)}$:

- ① **Distinct Real Poles:**

$$\begin{cases} m < n: & \text{Partial fractions on } \frac{H(z)}{z} \text{ with } K_i = (z - p_i) \frac{H(z)}{z} \Big|_{z = p_i} \\ m \ge n: & \text{Long division; note } \mathcal{Z}^{-1}(z^k) = \delta(n + k) \end{cases}$$

- ② **Complex Conjugate Poles on/near Circle:**

$$\begin{cases} \dfrac{z \sin(\omega_0 T)}{z^2 - 2z \cos(\omega_0 T) + 1} \longleftrightarrow \pm \sin(n \omega_0 T) u(n/ - n - 1) \\ \dfrac{z(z - \cos(\omega_0 T))}{z^2 - 2z \cos(\omega_0 T) + 1} \longleftrightarrow \pm \cos(n \omega_0 T) u(n/ - n - 1) \end{cases}$$

- ③ **Multiple Poles of Order k at $z = p$:**

$$H(z) = \sum_{r=1}^{k} \frac{C_r z}{(z - p)^r} + \frac{E(z)}{P(z)}$$

where coefficients are found via:

$$C_1 = \frac{X(z)}{z}(z-p)^k\bigg|_{z=p} = X_1(z)|_{z=p}, \quad C_2 = \frac{X_1(z)-C_1}{z-p}\bigg|_{z=p} = X_2(z)|_{z=p}, \quad C_3 = \frac{X_2(z)-C_2}{z-p}\bigg|_{z=p} = X_3(z)|_{z=p}, \dots$$

Inverse transform pair: $\mathcal{Z}^{-1}\left[\frac{z}{(z-1)^m}\right] = \pm\binom{n+m-1}{m-1}u(n/-n-1)$ (written $\pm C_n^m u(n/-n-1)$).

3. System Function $H(z)$ Properties

- ① **Invertibility:** $H(z)\cdot\tilde{H}(z)=1$
- ② **Causality:** For rational $H(z)$, the ROC is the exterior of a circle including $z=\infty$ ($|z|>r_{\max}$), and the numerator degree in z is $\leq$ the denominator degree.
- ③ **Stability:**
  1. **Stable $\Longleftrightarrow$ ROC contains the unit circle $|z|=1$:**

$$\sum_{n=-\infty}^{\infty}|h(n)|<\infty \Longleftrightarrow \sum_{n=-\infty}^{\infty}|h(n)z^{-n}|\bigg|_{|z|=1} < \infty \implies |H(z)|\Big|_{|z|=1}<\infty$$

  2. **Marginally Stable** $\Longleftrightarrow$ ROC has the unit circle $|z|=1$ as its boundary (with simple poles on $|z|=1$).
  3. **Unstable** $\Longleftrightarrow$ ROC does not contain the unit circle $|z|=1$.

Appendix A10: Bilateral z-Transform Properties

Property	Sequence $x(n)$	z-Transform $X(z)$	ROC
① Multiplication by n^m	$n^m x(n)$	$\left(-z\frac{d}{dz}\right)^m X(z)$	R
② Time Shifting (n-move)	$x(n-n_0)$	$z^{-n_0}X(z)$	R (may add or remove $0, \infty$)
③ Scaling in z-Domain	$z_0^n x(n)$	$X\left(\frac{z}{z_0}\right)$	$\lvert z_0\rvert R$
④ Time Reversal	$x(-n)$	$X(z^{-1})$	R^{-1}
⑤ Conjugation	$x^*(n)$	$X^*(z^*)$	R
⑥ Convolution	$x_1(n) * x_2(n)$	$X_1(z)\cdot X_2(z)$	At least $R_1 \cap R_2$
⑦ Initial Value Theorem	If $x(n)=x(n)u(n)$ (causal), then $x(0)=\lim_{z\to\infty} X(z)$, and generally $x(k)=\lim_{z\to\infty} z^k\left[X(z)-\sum_{r=0}^{k-1}x(r)z^{-r}\right]$		
⑧ Final Value Theorem	If $x(n)=x(n)u(n)$ and $X(z)$ is stable (or has at most a simple pole at $p=1$), then $x(\infty)=\lim_{z\to1}(z-1)X(z)$		
⑨ Correlation	$x_1(n) * x_2(-n)$	$X_1(z)\cdot X_2(z^{-1})$	At least $R_1 \cap R_2^{-1}$

Appendix A11: Discrete-Time Response Classification, All-Pass, and Minimum-Phase Systems

- ① **Transient, Steady-State, and Divergent Responses:**
  - **Transient Response:** $R(z)$ stable $\implies$ $H(z)$ and $E(z)$ are both stable.
  - **Steady-State Response:** $R(z)$ marginally stable $\implies$ $H(z)$ or $E(z)$ is marginally stable while the other is stable.
  - **Divergent Response:** $R(z)$ unstable $\implies$ $H(z)$ or $E(z)$ is unstable.
  - *Special Case:* Pole-zero cancellation between $H(z)$ and $E(z)$.
- ② **Natural vs. Forced Response in z-Domain:**
  - **Natural Response:** Produced by poles of $H(z)$.
  - **Forced Response:** Produced by poles of $E(z)$.
  - *Special Case:* When $H(z)$ and $E(z)$ share identical poles, the partial fraction terms of $R(z)$ whose pole orders match those of $H(z)$ constitute the natural response.
- ③ **Discrete-Time All-Pass and Minimum-Phase Systems:**
  - **First-order all-pass:** $H_1(z)=\frac{z^{-1}-z_p^*}{1-z_pz^{-1}}$; **Second-order all-pass:** $H_2(z)=\frac{z^{-1}-z_p^*}{1-z_pz^{-1}}\cdot\frac{z^{-1}-z_p}{1-z_p^*z^{-1}}$; **General all-pass:** $H_{ap}(z)=\prod_i H_{1i}(z)\prod_j H_{2j}(z)$.
  - **All-pass group delay property:** $-\frac{d}{d\omega}\varphi_{ap}(\omega)>0$.

- **Minimum-Phase System $H_{\min}(z)$:** All poles and zeros lie inside (or on) the unit circle $|z| \leq 1$.
- **Factorization:** Any stable, causal non-minimum-phase system can be decomposed as $H(z) = H_{ap}(z) \cdot H_{\min}(z)$.
- **Properties:** $-\frac{d}{d\omega}\varphi(\omega) > -\frac{d}{d\omega}\varphi_{\min}(\omega)$; and if $|H(e^{j\omega})| = |H_{\min}(e^{j\omega})|$, partial energy satisfies $\sum_{n=0}^{M} |h_{\min}(n)|^2 \geq \sum_{n=0}^{M} |h(n)|^2$ (written $\langle h_{\min}(n)|h_{\min}(n)\rangle \geq \langle h(n)|h(n)\rangle$).

Appendix A12: Unilateral z-Transform Properties

Definition: $X(z) = \sum_{n=0}^{\infty} x(n)z^{-n} = \sum_{n=-\infty}^{\infty} x(n)u(n)z^{-n}$, with inverse $x(n) = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}\,dz$.

- ① **Convolution:** When both $x_1(n)$ and $x_2(n)$ are causal, $x_1(n) * x_2(n) \to X_1(z)X_2(z)$.
- ② **Time Shifting (n-move, Advance & Delay):**

$$\mathcal{Z}[x(n+m)u(n)] = z^{m}\left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k}\right] \quad (m > 0)$$

$$\mathcal{Z}[x(n-m)u(n)] = z^{-m}\left[X(z) + \sum_{k=-m}^{-1} x(k)z^{-k}\right] \quad (m > 0)$$

IX. Discrete-Time Fourier Transform Domain (ω-Domain / DTFT)

1. Signal Representation

- ① **Direct Transform (DTFT):**

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-jn\omega} \quad (z\text{-Transform evaluated on the unit circle } |z|=1)$$

Relation to $X(z)$ based on $\mathrm{ROC}(X(z))$:

$$\begin{cases} \mathrm{ROC}(X(z)) \text{ contains } |z|=1: & X(e^{j\omega}) = X(z)|_{z=e^{j\omega}} \\ \mathrm{ROC}(X(z)) \text{ does not contain } |z|=1: & X(e^{j\omega}) \text{ does not exist} \\ \mathrm{ROC}(X(z)) \text{ has } |z|=1 \text{ as boundary}: & \text{If } |z|=1 \text{ has } k\text{-th order poles, } X(e^{j\omega}) \text{ includes } \delta^{(k-1)} \text{ impulses} \end{cases}$$

- ② **Inverse Transform (IDTFT):**

$$x(n) = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{jn\omega}\,d\omega$$

2. System Response & Spectral Density

Under zero-state conditions, $r(n) = e(n) * h(n)$, so:

$$R(e^{j\omega}) = E(e^{j\omega}) \cdot H(e^{j\omega}) \quad \text{if } \mathrm{ROC}(E) \text{ and } \mathrm{ROC}(H) \text{ contain } |z|=1.$$

- **Energy Sequences:**

$$\mathcal{E}_r(e^{j\omega}) = \mathcal{E}_e(e^{j\omega}) \cdot |H(e^{j\omega})|^2, \quad R_r(n) = R_e(n) * h(n) * h^*(-n) = R_e(n) * R_h(n)$$

- **Power Sequences:**

$$P_r(e^{j\omega}) = P_e(e^{j\omega}) \cdot |H(e^{j\omega})|^2, \quad R_r(n) = R_e(n) * h(n) * h^*(-n) = R_e(n) * R_h(n)$$

3. System Function $H(e^{j\omega})$ Properties

- ① **Invertibility:** $H(e^{j\omega}) \cdot \tilde{H}(e^{j\omega}) = 1$
- ② **Causality:** For rational $H(e^{j\omega})\big|_{e^{j\omega}=z}$, the ROC is the exterior of a circle containing $|z|=1$ (and $z=\infty$).
- ③ **Stability:** Always BIBO stable.

X. Discrete Fourier Series / Transform Domain (ω_0-Domain, $\omega_0 = \frac{2\pi}{N}$)

1. Signal Representation (DFS / DFT, where $W_N = e^{-j\frac{2\pi}{N}}$)

- ① **Inverse DFS / IDFT:**

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j(\frac{2\pi}{N})nk} = \frac{1}{N}\sum_{k=0}^{N-1} X(k)W_N^{-nk}$$

- ② **Direct DFS / DFT:**

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j(\frac{2\pi}{N})nk} = \sum_{n=0}^{N-1} x(n) W_N^{nk}$$

2. System Response via Periodic Extension (Linear Convolution via Circular/DFS)

Under zero-state conditions, $r(n) = e(n) * h(n)$, so:

$$R(k) = E(k) \cdot H(k)$$

Suppose $e(n)$ has finite length N_1 and $h(n)$ has finite length N_2; then the linear convolution $r(n) = e(n) * h(n)$ has length $N_3 = N_1 + N_2 - 1$. Periodically extending $e(n), h(n), r(n)$ with period $N \geq N_3$ yields $e_N(n), h_N(n), r_N(n)$, whose DFS with fundamental frequency $\omega_0 = \frac{2\pi}{N}$ give $E(k), H(k), R(k)$.

3. System Function $H(k)$ Properties

- ① **Invertibility:** $H(k) \cdot \tilde{H}(k) = 1$, for $N > N_h + N_{\tilde{h}}$.
- ② **Causality:** For rational $H(k)\big|_{e^{j\frac{2\pi}{N}k}=z}$, the ROC is the exterior of $|z| = 1$.
- ③ **Stability:** Always stable (since $h(n)$ is a periodic extension of a finite-duration impulse response).

Appendix A13: DTFT of Periodic Sequences

Let $x(n)$ be periodic with period N. Then:

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi}{N}nk}, \quad X(e^{j\omega}) = \frac{2\pi}{N}\sum_{r=-\infty}^{\infty}\sum_{k=0}^{N-1} X(k)\delta\left(\omega - \frac{2\pi}{N}k - 2\pi r\right), \quad X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi}{N}nk} = X_N(e^{j\omega})\Big|_{\omega=\frac{2\pi}{N}k}$$

The power spectral density is
$P(e^{j\omega}) = \mathcal{F}[R(m)] = \frac{2\pi}{N^2}\sum_{r=-\infty}^{\infty}\sum_{k=0}^{N-1}|X(k)|^2\delta\left(\omega - \frac{2\pi}{N}k - 2\pi r\right).$

Appendix A14: DTFT Properties

1. **Time Shifting (n-move):** $x(n-n_0) \to e^{-j\omega n_0}X(e^{j\omega})$
2. **Frequency Shifting (ω-move):** $e^{j\omega_0 n}x(n) \to X\left(e^{j(\omega-\omega_0)}\right)$
3. **Frequency Differentiation:** $nx(n) \to j\frac{d}{d\omega}X(e^{j\omega})$
4. **Convolution & Multiplication:**

$$x(n) * y(n) \to X(e^{j\omega})\cdot Y(e^{j\omega}), \quad x(n)\cdot y(n) \to \frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y\left(e^{j(\omega-\theta)}\right)d\theta$$

5. **Correlation:** $R_{xy}(m) = x(m) * y^*(-m) \to X(e^{j\omega})Y^*(e^{j\omega})$

Appendix A15: Reconstructing DTFT Spectrum from DFS / DFT

Let $x(n)$ be a finite-length sequence of length N_0 (supported on $0 \le n \le N_0 - 1$).
Periodically extending $x(n)$ with period $N \ge N_0$ gives $x_N(n)$:

- DTFT of $x(n)$: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-jn\omega}$
- DFS of $x_N(n)$: $X(k) = \sum_{n=0}^{N-1} x_N(n)e^{-j(\frac{2\pi}{N})nk}$

Since $x(n) = x_N(n)[u(n) - u(n-N)]$, periodic convolution in the frequency domain yields the exact interpolation formula:

$$X(e^{j\omega}) = \frac{1}{2\pi}\left[\frac{2\pi}{N}\sum_{r=-\infty}^{\infty}\sum_{k=0}^{N-1} X(k)\delta\left(\omega - \frac{2\pi}{N}k - 2\pi r\right)\right] * \frac{1-e^{-jN\omega}}{1-e^{-j\omega}} = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\frac{1-e^{-jN\omega}}{1-e^{-j(\omega-\frac{2\pi}{N}k)}}$$

XI. Sampling Theory (Time & Frequency Domains)

1. Time-Domain Sampling of Continuous Aperiodic Signals

- ① **Discrete-Time Sequence Representation ($f(t) \to f(nT) = x(n)$):**

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n)e^{-jn\omega} = \sum_{n=-\infty}^{\infty} f(nT)e^{-jn\omega} = \sum_{n=-\infty}^{\infty} \left[\frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\Omega)e^{j\Omega nT}\,d\Omega\right]e^{-jn\omega} \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\Omega)\,d\Omega \sum_{n=-\infty}^{\infty} e^{jn(\Omega T-\omega)} = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(j\Omega)\,d\Omega \cdot 2\pi\sum_{r=-\infty}^{\infty}\delta(\omega-\Omega T-2\pi r) \\ &= \frac{1}{T}\int_{-\infty}^{\infty} F(j\Omega)\,d\Omega \sum_{r=-\infty}^{\infty}\delta\left(\Omega-\frac{\omega}{T}-\frac{2\pi}{T}r\right) = \frac{1}{T}\sum_{r=-\infty}^{\infty} F\left(j\frac{\omega}{T}+j\frac{2\pi}{T}r\right) \end{aligned}$$

- ② **Impulse-Train Modulated Representation ($f(t) \to f_s(t) = f(t)\sum_{n=-\infty}^{\infty}\delta(t-nT)$):**

$$F_s(j\Omega) = \mathcal{F}\left[f(t)\cdot\sum_{n=-\infty}^{\infty}\delta(t-nT)\right] = \frac{1}{2\pi}F(j\Omega) * \frac{2\pi}{T}\sum_{r=-\infty}^{\infty}\delta\left(\Omega+\frac{2\pi}{T}r\right) = \frac{1}{T}\sum_{r=-\infty}^{\infty} F\left(j\Omega+j\frac{2\pi}{T}r\right)$$

2. Time-Domain Sampling of Continuous Periodic Signals

Assume the signal period T is an integer multiple N of the sampling interval T_s ($T = NT_s$):

- ① **Discrete Periodic Sequence ($f(t) \to f(nT_s) = x(n)$):**

$$\begin{aligned} X(k) &= \sum_{n=0}^{N-1} x(n)e^{-j(\frac{2\pi}{N})nk} = \sum_{n=0}^{N-1} f(nT_s)e^{-j(\frac{2\pi}{N})nk} = \sum_{n=0}^{N-1}\sum_{m=-\infty}^{\infty} F_m e^{jm(\frac{2\pi}{T})nT_s}e^{-j(\frac{2\pi}{N})nk} \\ &= \sum_{m=-\infty}^{\infty} F_m \sum_{n=0}^{N-1} e^{jn(\frac{2\pi}{N}m-\frac{2\pi}{N}k)} = \sum_{m=-\infty}^{\infty} F_m\cdot N\sum_{r=-\infty}^{\infty}\delta(m-k-rN) = N\sum_{r=-\infty}^{\infty} F_{k+rN} \end{aligned}$$

- ② **Impulse-Train Sampled Periodic Signal ($f(t) \to f(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s)$):**

$$\begin{aligned} F_m &= \frac{1}{T}\int_T \left[f(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s)\right]e^{-jm(\frac{2\pi}{T})t}\,dt = \frac{1}{T}\sum_{\ell=-\infty}^{\infty} F_\ell \sum_{n=-\infty}^{\infty}\int_T \delta(t-nT_s)e^{j\frac{2\pi}{T}nT_s(\ell-m)}\,dt \\ &= \frac{1}{T}\sum_{\ell=-\infty}^{\infty} F_\ell\cdot 2\pi\sum_{r=-\infty}^{\infty}\delta\left[\frac{2\pi}{N}(\ell-m)-2\pi r\right] = \frac{2\pi}{T}\sum_{r=-\infty}^{\infty}\sum_{\ell=-\infty}^{\infty} F_\ell\frac{N}{2\pi}\delta(\ell-m-rN) = \frac{1}{T_s}\sum_{r=-\infty}^{\infty} F_{m+rN} \end{aligned}$$

3. Frequency-Domain Sampling of Continuous Aperiodic Signals

- ① **Fourier Series from Spectral Samples ($F(j\omega) \to F(jn\Omega) = F_n$):**

$$\begin{aligned} f_s(t) &= \sum_{n=-\infty}^{\infty} F_n e^{jn\Omega t} = \sum_{n=-\infty}^{\infty}\left[\int_{-\infty}^{\infty} f(\tau)e^{-jn\Omega\tau}\,d\tau\right]e^{jn\Omega t} = \int_{-\infty}^{\infty} f(\tau)\,d\tau\sum_{n=-\infty}^{\infty} e^{jn\Omega(t-\tau)} \\ &= \int_{-\infty}^{\infty} f(\tau)\,d\tau\cdot 2\pi\sum_{r=-\infty}^{\infty}\delta[\Omega(t-\tau)-2\pi r] = \frac{2\pi}{\Omega}\int_{-\infty}^{\infty} f(\tau)\,d\tau\sum_{r=-\infty}^{\infty}\delta\left(\tau-t+\frac{2\pi}{\Omega}r\right) = \frac{2\pi}{\Omega}\sum_{r=-\infty}^{\infty} f\left(t-\frac{2\pi}{\Omega}r\right) \end{aligned}$$

- ② **Impulse-Train Sampled Spectrum ($F(j\omega) \to F(j\omega)\sum_{n=-\infty}^{\infty}\delta(\omega-n\Omega)$):**

$$\tilde{f}_s(t) = \mathcal{F}^{-1}\left[F(j\omega)\cdot\sum_{n=-\infty}^{\infty}\delta(\omega-n\Omega)\right] = f(t) * \frac{2\pi}{\Omega}\sum_{m=-\infty}^{\infty}\delta\left(t-m\frac{2\pi}{\Omega}\right) = \frac{2\pi}{\Omega}\sum_{m=-\infty}^{\infty} f\left(t-\frac{2\pi}{\Omega}m\right)$$

Appendix A16: Fast Fourier Transform (FFT) Algorithms

Let $W_N = e^{-j\frac{2\pi}{N}}$. Using $W_N^2 = W_{N/2}$, $W_N^4 = W_{N/4}$, and $W_N^{N/2} = -1$:

① Radix-2 Decimation-in-Time (DIT) FFT

Split $x(n)$ into even ($n = 2r$) and odd ($n = 2r+1$) indices:

$$X(k) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_N^{2rk} + \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_N^{(2r+1)k} = \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk} + W_N^k \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{N/2}^{rk}$$

$$X\left(k+\frac{N}{2}\right) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk} + W_N^{k+\frac{N}{2}} \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{N/2}^{rk} = \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk} - W_N^k \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{N/2}^{rk}$$

② Radix-2 Decimation-in-Frequency (DIF) FFT

Split the summation interval into first half ($0 \le n < N/2$) and second half ($n + N/2$), then evaluate at even ($k = 2r$) and odd ($k = 2r+1$) frequencies:

$$X(2r) = \sum_{n=0}^{N-1} x(n) W_N^{2rn} = \sum_{n=0}^{\frac{N}{2}-1} x(n) W_{N/2}^{rn} + \sum_{n=0}^{\frac{N}{2}-1} x\left(n+\frac{N}{2}\right) W_{N/2}^{r(n+\frac{N}{2})} = \sum_{n=0}^{\frac{N}{2}-1} \left[x(n) + x\left(n+\frac{N}{2}\right)\right] W_{N/2}^{rn}$$

$$X(2r+1) = \sum_{n=0}^{N-1} x(n) W_N^{(2r+1)n} = \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^n W_{N/2}^{rn} + \sum_{n=0}^{\frac{N}{2}-1} x\left(n+\frac{N}{2}\right) W_N^{(2r+1)(n+\frac{N}{2})} = \sum_{n=0}^{\frac{N}{2}-1} \left\{\left[x(n) - x\left(n+\frac{N}{2}\right)\right] W_N^n\right\} W_{N/2}^{rn}$$

③ Radix-4 Decimation-in-Time (DIT) FFT

Decompose into 4 subsequences $x(4r+i)$ ($i = 0, 1, 2, 3$):

$$\begin{aligned} X(k) &= \sum_{r=0}^{\frac{N}{4}-1} x(4r) W_N^{4rk} + \sum_{r=0}^{\frac{N}{4}-1} x(4r+1) W_N^{(4r+1)k} + \sum_{r=0}^{\frac{N}{4}-1} x(4r+2) W_N^{(4r+2)k} + \sum_{r=0}^{\frac{N}{4}-1} x(4r+3) W_N^{(4r+3)k} \\ &= \sum_{r=0}^{\frac{N}{4}-1} x(4r) W_{N/4}^{rk} + W_N^k \sum_{r=0}^{\frac{N}{4}-1} x(4r+1) W_{N/4}^{rk} + W_N^{2k} \sum_{r=0}^{\frac{N}{4}-1} x(4r+2) W_{N/4}^{rk} + W_N^{3k} \sum_{r=0}^{\frac{N}{4}-1} x(4r+3) W_{N/4}^{rk} \end{aligned}$$

And similarly for $X\left(k+\frac{N}{4}\right), X\left(k+\frac{N}{2}\right), X\left(k+\frac{3N}{4}\right)$ using $W_N^{m(k+qN/4)} = (-j)^{mq} W_N^{mk}$.

④ Radix-4 Decimation-in-Frequency (DIF) FFT

Split the input into 4 quarters $x\left(n+i\frac{N}{4}\right)$ ($i = 0, 1, 2, 3$):

$$\begin{aligned} X(4r) &= \sum_{n=0}^{\frac{N}{4}-1} x(n) W_N^{4nr} + \sum_{n=\frac{N}{4}}^{\frac{N}{2}-1} x(n) W_N^{4nr} + \sum_{n=\frac{N}{2}}^{\frac{3N}{4}-1} x(n) W_N^{4nr} + \sum_{n=\frac{3N}{4}}^{N-1} x(n) W_N^{4nr} \\ &= \sum_{n=0}^{\frac{N}{4}-1} \left[x(n) + x\left(n+\frac{N}{4}\right) + x\left(n+\frac{N}{2}\right) + x\left(n+\frac{3N}{4}\right)\right] W_{N/4}^{rn} \end{aligned}$$

With analogous expressions for $X(4r+1), X(4r+2), X(4r+3)$.

⑤ Split-Radix DIT FFT & ⑥ Split-Radix DIF FFT

Combines Radix-2 for even indices ($n = 2r$) and Radix-4 for odd indices ($n = 4\ell+1, 4\ell+3$):

$$
\begin{aligned}
X(k) &= \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_N^{2rk} + \sum_{\ell=0}^{\frac{N}{4}-1} x(4\ell+1) W_N^{(4\ell+1)k} + \sum_{\ell=0}^{\frac{N}{4}-1} x(4\ell+3) W_N^{(4\ell+3)k} \\
&= \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk} + W_N^k \sum_{\ell=0}^{\frac{N}{4}-1} x(4\ell+1) W_{N/4}^{\ell k} + W_N^{3k} \sum_{\ell=0}^{\frac{N}{4}-1} x(4\ell+3) W_{N/4}^{\ell k}
\end{aligned}
$$

(And dual formulation for ⑥ Split-Radix DIF FFT.)

XI (cont.). Frequency-Domain Sampling & Multirate Signal Processing

4. Frequency-Domain Sampling of Periodic Spectrum $X(e^{j\omega})$

Assume the frequency period 2π is an integer multiple N of the frequency sampling interval Ω_s ($2\pi = N\Omega_s$, i.e., $\Omega_s = \frac{2\pi}{N}$):

- ① **IDFS from DTFT Samples ($X(e^{j\omega}) \to X(e^{jk\Omega_s}) = X(k)$):**

$$x_s(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j(\frac{2\pi}{N})nk} = \frac{1}{N}\sum_{k=0}^{N-1}\left[\sum_{m=-\infty}^{\infty} x(m)e^{-jmk\Omega_s}\right]e^{j\Omega_s nk}$$
$$= \frac{1}{N}\sum_{m=-\infty}^{\infty} x(m)\sum_{k=0}^{N-1} e^{j\Omega_s k(n-m)} = \sum_{m=-\infty}^{\infty} x(m)\sum_{r=-\infty}^{\infty}\delta(n-m+rN) = \sum_{r=-\infty}^{\infty} x(n+rN)$$

- ② **Impulse-Train Sampled DTFT ($X(e^{j\omega}) \to X(e^{j\omega})\sum_{k=-\infty}^{\infty}\delta(\omega - k\Omega_s)$):**

$$\tilde{x}_s(n) = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\sum_{k=-\infty}^{\infty}\delta(\omega-k\Omega_s)e^{j\omega n}\,d\omega = \frac{1}{2\pi}\sum_{m=-\infty}^{\infty} x(m)\sum_{k=-\infty}^{\infty} e^{jk\Omega_s(n-m)}$$
$$= \sum_{m=-\infty}^{\infty} x(m)\sum_{r=-\infty}^{\infty}\delta[\Omega_s(n-m)+2\pi r] = \sum_{m=-\infty}^{\infty} x(m)\frac{1}{\Omega_s}\sum_{r=-\infty}^{\infty}\delta(m-n-rN) = \frac{1}{\Omega_s}\sum_{r=-\infty}^{\infty} x(n+rN)$$

5. Downsampling (Decimation) by Integer Factor M

Anti-aliasing filter followed by decimation: $x(n) \xrightarrow{h(n)} w(n) \xrightarrow{\downarrow M} y(n)$.

$$\begin{cases} w(n) = \sum_k h(k)x(n-k) \\ y(n) = w(Mn) \end{cases} \implies y(n) = \sum_k h(k)x(Mn-k) = \sum_k h(Mn-k)x(k)$$

Using the identity $\frac{1}{M}\sum_{\ell=0}^{M-1} e^{j\frac{2\pi}{M}\ell n} = \begin{cases} 1, & n = mM \\ 0, & \text{otherwise} \end{cases}$:

$$Y(z) = \sum_m y(m)z^{-m} = \sum_m w(Mm)z^{-m} = \sum_{n=-\infty}^{\infty} w(n)z^{-n/M}\frac{1}{M}\sum_{\ell=0}^{M-1} e^{j\frac{2\pi}{M}\ell n} = \frac{1}{M}\sum_{\ell=0}^{M-1} W\left(e^{-j\frac{2\pi}{M}\ell}z^{1/M}\right)$$

Evaluating on the unit circle ($z = e^{j\omega}$):

$$Y(e^{j\omega}) = \frac{1}{M}\sum_{\ell=0}^{M-1} H\left(e^{j(\frac{\omega-2\pi\ell}{M})}\right)X\left(e^{j(\frac{\omega-2\pi\ell}{M})}\right) \implies \frac{1}{M}X\left(e^{j\frac{\omega}{M}}\right), \quad |\omega| \le \pi \quad \text{(with ideal lowpass } H\text{)}$$

6. Upsampling (Interpolation) by Integer Factor L

Zero-insertion followed by anti-imaging filter: $x(n) \xrightarrow{\uparrow L} w(n) \xrightarrow{h(n)} y(n)$.

$$\begin{cases} w(n) = x(n/L) \quad (\text{for } n = mL, \text{ else } 0) \\ y(n) = \sum_k h(n-k)w(k) \end{cases} \implies y(n) = \sum_k h(n-k)x(k/L) = \sum_k h(n-kL)x(k)$$

In the z-domain and frequency domain:

$$Y(z) = H(z)W(z) = H(z)\sum_m w(m)z^{-m} = H(z)\sum_m x(m)z^{-mL} = H(z)X(z^L)$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega L}) \implies LX(e^{j\omega L}), \quad |\omega| \le \pi \quad \text{(with ideal lowpass } H \text{ of gain } L\text{)}$$

7. Rational Sampling Rate Conversion by L/M (Interpolate by L, then Decimate by M)

$$y(n) = \sum_{r=-\infty}^{\infty} h(Mn - Lr)x(r)$$

$$Y(e^{j\omega}) = \frac{1}{M}\sum_{\ell=0}^{M-1} H\left(e^{j\left(\frac{\omega-2\pi\ell}{M}\right)}\right) X\left(e^{jL\left(\frac{\omega-2\pi\ell}{M}\right)}\right) \implies \frac{L}{M} X\left(e^{j\frac{L}{M}\omega}\right), \quad |\omega| \leq \min\left\{\pi, \pi\frac{M}{L}\right\}$$

Header Note: **Minimum-Phase System:** Among all systems sharing the same magnitude response, the minimum-phase system has the minimum group delay (and minimum phase lag).

XII. Analog and Digital Filters

1. Analog Filter Synthesis (Doubly-Terminated Lossless LC Network — Darlington Synthesis)

Consider a source E_s with internal resistance R_s driving a two-port lossless LC network (ports 1-1′ and 2-2′, input impedance $Z_{11}(s)$, input voltage V_i, input current I_1) terminated in load resistance R_L (output voltage V_o, current I_2).

- **Problem Statement:** Given that $H_a(s) = \frac{V_o(s)}{E_s(s)}$ is stable, causal, minimum-phase, and conjugate-symmetric, and given $|H_a(j\Omega)|^2$, determine the driving-point impedance $Z_{11}(s)$ and synthesize the LC ladder network.
- **Solution Procedure:**
  - From $|H_a(j\Omega)|^2\big|_{j\Omega=s} = H_a(s)H_a(-s)$ and the stability/minimum-phase conditions, identify $H_a(s)$.
  - **Available (Rated) Source Power:** $P_m = \frac{1}{2}\frac{|E_s|^2}{4R_s}$
  - **Load Power:** $P_L = \frac{1}{2}\frac{|V_o|^2}{R_L}$
  - **Transducer Power Gain (Transmission Function):** $|K|^2 = \frac{P_L}{P_m} = \frac{4R_s}{R_L}\left|\frac{V_o}{E_s}\right|^2 = \frac{4R_s}{R_L}|H_a(j\Omega)|^2$
  - **Reflection Coefficient:** $|\rho|^2 = 1 - \frac{4R_s}{R_L}|H_a(s)|^2$
  - Since the LC network is lossless, input power at port 1-1′ equals load power ($P_1 = P_2 = P_L$):

$$\frac{1}{2}\mathrm{Re}[Z_{11}(j\Omega)]\,|I_1(j\Omega)|^2 = \frac{1}{2}\frac{|V_o(j\Omega)|^2}{R_L} \implies \left|\frac{V_o}{I_1}\right|^2 = R_L \cdot \mathrm{Re}[Z_{11}(s)]$$

  - Also, since $\frac{E_s}{I_1} = R_s + Z_{11}(s)$:

$$|H_a(s)|^2 = \left|\frac{V_o}{I_1}\right|^2\left|\frac{I_1}{E_s}\right|^2 = \frac{R_L \cdot \mathrm{Re}[Z_{11}(s)]}{|R_s + Z_{11}(s)|^2}$$

$$|\rho(s)|^2 = \frac{|Z_{11} - R_s|^2}{|Z_{11} + R_s|^2} \implies \rho(s) = \pm\frac{Z_{11} - R_s}{Z_{11} + R_s} \implies Z_{11}(s) = R_s\frac{1 \pm \rho(s)}{1 \mp \rho(s)}$$

- **Network Realization Methods from $Z_{11}(s)$:**
  - ① **Foster Canonical Forms:** Series form ($Z_{11}(s) = \sum \cdots$ partial fractions) and Parallel form ($Y_{11}(s) = \sum \cdots$).
  - ② **Cauer Canonical (Ladder) Forms:** T-type (s continued fraction expansion around ∞) and Π-type ($\frac{1}{s}$ continued fraction expansion around 0).

2. Analog Filter Frequency Transformations (Normalized Prototype s', L', C')

Filter Type	Complex Frequency Mapping $s' \leftarrow$	Inductor $s'L' \leftarrow$	Capacitor $\frac{1}{s'C'} \leftarrow$
Prototype	s'	$s'L'$	$\frac{1}{s'C'}$
Lowpass (LP)	$\frac{s}{\Omega_c}$	$s\frac{R_s}{\Omega_c}L'$	$\frac{\Omega_c R_s}{sC'}$
Highpass (HP)	$\frac{\Omega_c}{s}$	$\frac{1}{s\frac{1}{\Omega_c R_s L'}}$	$s\frac{R_s}{C'\Omega_c}$
Bandpass (BP)	$\frac{s^2+\Omega_0^2}{s\Omega_r}$	$s\frac{R_s L'}{\Omega_r} + \frac{\Omega_r}{s\Omega_0^2 R_s L'}$ (Series LC)	$\left(s\frac{\Omega_r R_s}{\Omega_0^2 C'}\right) \parallel \left(\frac{C'}{s\Omega_r R_s}\right)$ (Parallel LC)
Bandstop (BS)	$\frac{s\Omega_r}{s^2+\Omega_0^2}$	$\left(s\frac{\Omega_r R_s L'}{\Omega_0^2}\right) \parallel \left(\frac{1}{s\Omega_r R_s L'}\right)$ (Parallel LC)	$s\frac{R_s}{\Omega_r C'} + \frac{\Omega_r C'}{s\Omega_0^2 R_s}$ (Series LC)

(where $\Omega_r = \Omega_2 - \Omega_1$ is the bandwidth and $\Omega_0^2 = \Omega_1\Omega_2$ is the center frequency squared.)

3. IIR Digital Filter Structures

$$H(z) = \frac{\sum_{r=0}^{M} b_r z^{-r}}{1 + \sum_{k=1}^{N} a_k z^{-k}}, \quad y(n) = \sum_{r=0}^{M} b_r x(n-r) - \sum_{k=1}^{N} a_k y(n-k)$$

- ① **Direct Form Structures (Direct Form I & Canonical Direct Form II):**
  - Direct Form II uses the minimum number of delay elements ($\max(M, N)$).
  - Poles and zeros are difficult to tune independently; highly sensitive to finite word-length (coefficient quantization) effects.
- ② **Cascade Form Structure:**
  - Factored into second-order sections (biquads). Uses minimum delay elements; zeros and poles can be tuned independently.
  - **Pairing & Ordering Rules:** Pair poles with their closest zeros; order sections by decreasing gain (from the perspective of minimizing quantization noise) or by decreasing pole distance to the unit circle (from the perspective of signal attenuation / preventing overflow).
- ③ **Parallel Form Structure:** (Continued on Page 19)

Appendix A17: Butterworth Analog Filter Design & Synthesis

1. Magnitude-Squared Response & Pole Distribution

Given the N-th order Butterworth magnitude-squared specification:

$$|H_a(j\Omega)|^2 = \left(\frac{1}{\sqrt{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}}}\right)^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}} \implies |H_a(s)|^2 = H_a(s)H_a(-s) = \frac{(j\Omega_c)^{2N}}{s^{2N}+(j\Omega_c)^{2N}}$$

The $2N$ poles lie on a circle of radius Ω_c in the s-plane:

$$s_k = \Omega_c\, e^{j\left[\frac{1}{2N}(2k-1)\pi+\frac{\pi}{2}\right]}, \quad k = 1, 2, \ldots, 2N$$

2. Stable Transfer Function $H_a(s)$

Selecting all N left-half-plane (LHP) poles ($k = 1, 2, \ldots, N$) to form a stable and causal transfer function $H_a(s)$:

$$H_a(s) = \frac{\Omega_c^N}{\prod_{k=1}^{N}(s-s_k)} = \begin{cases} \Omega_c^N \Big/ \left\{(s+\Omega_c)\displaystyle\prod_{k=1}^{(N-1)/2}\left[s^2 - 2\Omega_c\cos\left(\frac{2k-1}{2N}\pi+\frac{\pi}{2}\right)s+\Omega_c^2\right]\right\}, & N \text{ odd} \\ \Omega_c^N \Big/ \displaystyle\prod_{k=1}^{N/2}\left[s^2 - 2\Omega_c\cos\left(\frac{2k-1}{2N}\pi+\frac{\pi}{2}\right)s+\Omega_c^2\right], & N \text{ even} \end{cases}$$

With normalized frequency $s' = s/\Omega_c$, the normalized Butterworth polynomial $B_N(s')$ gives:

$$H_a(s') = \frac{1}{B_N(s')} = \begin{cases} 1 \Big/ \left\{(s'+1)\displaystyle\prod_{k=1}^{(N-1)/2}\left[s'^2 - 2\cos\left(\frac{2k-1}{2N}\pi+\frac{\pi}{2}\right)s'+1\right]\right\}, & N \text{ odd} \\ 1 \Big/ \displaystyle\prod_{k=1}^{N/2}\left[s'^2 - 2\cos\left(\frac{2k-1}{2N}\pi+\frac{\pi}{2}\right)s'+1\right], & N \text{ even} \end{cases}$$

3. Darlington Impedance Synthesis & Element Denormalization

Given passband and stopband specifications, solve for N and Ω_c to determine $H_a(s)$.
Typically, source and load resistances are matched ($R_s = R_L$), so $H_a(j0) = \frac{1}{2}$. Then:

$$\rho(s')\rho(-s') = 1 - \frac{4R_s}{R_L}\cdot\frac{1/2}{\sqrt{1+(s'/j)^{2N}}}\cdot\frac{1/2}{\sqrt{1+(-s'/j)^{2N}}} = 1 - \frac{1}{1+(-1)^N(s')^{2N}} = \frac{(s')^N(-s')^N}{B_N(s')B_N(-s')}$$

Therefore, the reflection coefficient and driving-point impedance are:

$$\rho(s') = \frac{(s')^N}{B_N(s')} \implies Z_{11}(s') = R_s\frac{1\pm\rho(s')}{1\mp\rho(s')} = R_s\frac{B_N(s')\pm(s')^N}{B_N(s')\mp(s')^N}$$

Normalized with respect to source resistance R_s:

$$\frac{Z_{11}(s')}{R_s} = \frac{B_N(s')\pm(s')^N}{B_N(s')\mp(s')^N}$$

Denormalizing to actual physical inductance, capacitance, and resistance values:

$$\begin{cases} L = \dfrac{R_s}{\Omega_c} L' \\ C = \dfrac{1}{\Omega_c R_s} C' \\ R_L = R_s \end{cases}$$

XII (cont.). IIR Design Methods & FIR Filter Structures

3 (cont.). Parallel Form IIR Structure

Realized via partial fraction expansion of $H(z)$ into parallel first- and second-order sections:

- High computational speed (parallel execution); poles can be tuned independently; quantization/arithmetic errors in individual parallel branches do not accumulate across sections.
- Zeros are difficult to tune directly.

4. IIR Digital Filter Design Methods

① Impulse Invariance Method:

- **Principle:** Sample the analog impulse response: $h(nT_s) = h_a(t)|_{t=nT_s}$ (or $h(n) = T_s h_a(nT_s)$).
- **Mapping Result:**

$$\frac{1}{s-s_k} \Longrightarrow \frac{1}{1-e^{s_k T_s} z^{-1}}, \quad H(e^{j\omega}) = \frac{1}{T_s} \sum_{r=-\infty}^{\infty} H_a\left(j\frac{\omega}{T_s} + j\frac{2\pi}{T_s} r\right)$$

- **Conclusion:** Suffers from frequency aliasing; high-frequency response is distorted (unsuitable for highpass or bandstop filters).

② Bilinear Transformation Method:

- **Principle:** Two-step conformal mapping ($s \to s_1$-plane via tanh, then $s_1 \to z$-plane):
  - Step 1 ($s \to s_1$-plane, compressed strip $[-\pi/T_s, \pi/T_s]$):

$$\begin{cases} s = \frac{2}{T_s} \frac{1-e^{-s_1 T_s}}{1+e^{-s_1 T_s}} \\ \Omega = \frac{2}{T_s} \tan\left(\frac{\Omega_1 T_s}{2}\right) \Longleftrightarrow \Omega_1 = \frac{2}{T_s} \arctan\left(\frac{\Omega T_s}{2}\right) \end{cases}$$

  - Step 2 ($s_1 \to z$-plane):

$$\begin{cases} z = e^{s_1 T_s} \\ \omega = \Omega_1 T_s \end{cases}$$

- **Combined Mapping Result:**

$$s = \frac{2}{T_s} \frac{1-z^{-1}}{1+z^{-1}}, \quad \Omega = \frac{2}{T_s} \tan\left(\frac{\omega}{2}\right)$$

- **Conclusion:** Completely eliminates aliasing (one-to-one mapping of $j\Omega$-axis to unit circle), but introduces nonlinear frequency warping distortion (requires pre-warping at critical frequencies).

5. FIR Digital Filter Structures

$$H(z) = \sum_{n=0}^{N-1} h(n) z^{-n}, \quad y(n) = \sum_{r=0}^{N-1} h(r) x(n-r)$$

- ① **Direct Form Structure (Transversal / Tapped-Delay-Line Filter):** Delay line z^{-1} with tap weights $h(0), h(1), \ldots, h(N-1)$. Zeros are difficult to control directly.
- ② **Cascade Form Structure:** Product of second-order sections $(1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2})$. Zeros can be tuned independently.
- ③ **Fast Convolution Structure (Overlap-Add / Overlap-Save):** $x(n) \to \text{FFT} \to \times H(k) \to \text{IDFT} \to y(n)$. High computational speed for long N.
- ④ **Frequency Sampling Structure (Comb Filter + Parallel Resonators):**

$$H(z) = \sum_{n=0}^{N-1} h(n) z^{-n} = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{k=0}^{N-1} H(k) W_N^{-nk} \right) z^{-n} = \frac{1}{N} (1 - z^{-N}) \sum_{k=0}^{N-1} \frac{H(k)}{1 - W_N^{-k} z^{-1}}$$

- ⑤ **Linear-Phase FIR Structures:** For a linear-phase filter $H(e^{j\omega}) = |H(e^{j\omega})| e^{-j(\beta + \alpha\omega)}$ (or $\hat{H}(e^{j\omega}) e^{-j(\beta+\alpha\omega)}$):

$$\begin{cases} \text{Even Symmetry (Type I \& II):} & \beta = 0, \quad \alpha = \frac{N-1}{2}, \quad h(n) = h(N-1-n) \\ \text{Odd / Anti-Symmetry (Type III \& IV):} & \beta = \pm\frac{\pi}{2}, \quad \alpha = \frac{N-1}{2}, \quad h(n) = -h(N-1-n) \end{cases}$$

  1) **Direct Form Linear-Phase Realization:** Folds the delay line to add symmetric/anti-symmetric taps $x(n-k) \pm x(n-(N-1-k))$ before multiplying by $h(k)$, halving the number of multipliers (detailed for all 4 types on Page 21).

Appendix A18: Finite Word-Length Quantization Effects & Electronic Circuit Noise

1. Quantization Error Statistical Models (Step Size $q = 2^{-b}$)

Error Type	Distribution	Mean μ_e	Variance σ_e^2
Truncation Error (Two's Complement)	$U(-q, 0)$	$-\frac{q}{2}$	$\frac{q^2}{12} = \frac{1}{12} \times 2^{-2b}$
Rounding Error	$U\left(-\frac{q}{2}, \frac{q}{2}\right)$	0	$\frac{q^2}{12} = \frac{1}{12} \times 2^{-2b}$

Generally, two's complement fixed-point rounding is used to analyze quantization noise. The Signal-to-Quantization-Noise Ratio (SQNR) is:

$$\frac{S}{N} = 10 \lg \frac{\sigma_x^2}{\sigma_e^2} = 10 \lg \left(\frac{\sigma_x^2}{\frac{1}{12} \times 2^{-2b}} \right) = \left(6.02b + 10.79 + 10 \lg \sigma_x^2\right) \text{ dB}$$

2. Quantization Noise Propagation & Dynamic Range Scaling

- By the theory of random processes through linear systems, the quantization noise variance at the output of a digital filter with impulse response $h(m)$ from noise source to output is:

$$\sigma_f^2 = \sigma_e^2 \sum_{m=0}^{\infty} h^2(m) = \sigma_e^2 \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 \, d\omega$$

- For multiple independent quantization noise sources ($i = 1, \ldots, N$), superposition holds at the output:

$$\sigma_{\text{total}}^2 = \sigma_e^2 \sum_{i=1}^{N} \sum_{m=0}^{\infty} h_i^2(m)$$

- **Scaling to Prevent Overflow:** The above noise analysis is valid only when no addition overflow occurs at any internal node. To prevent fixed-point overflow, every internal node variable $w(n)$ must satisfy $|w(n)| \leq 1$. Suppose the input signal is bounded by $|x(n)| \leq 1$, and let $f(n)$ denote the unit impulse response from the input to node variable $w(n)$:

$$w(n) = f(n) * x(n) = \sum_{m=0}^{\infty} f(m) x(n-m)$$

  We scale the input to node $w(n)$ by dividing by β (and multiply by β at the output), where the scaling factor β is chosen according to:

$$\begin{cases} \ell_1\text{-norm criterion (strict):} & \beta = \sum_{n=0}^{\infty} |f(n)| = \|f\|_1 \\ \ell_2\text{-norm criterion (energy/RMS):} & \beta = \sqrt{\sum_{n=0}^{\infty} f^2(n)} = \|f\|_2 \end{cases}$$

  Trade-off: After introducing the compression scaling factor β, for the same input signal, the output quantization noise increases (because scaling requires multiplying the output or internal branches by β, amplifying the noise sources).

3. Electronic Device Noise Power Spectral Densities (PSD)

- **Resistor Thermal (Johnson–Nyquist) Noise:**

$$W_v(f) = 4kTR(f), \quad W_i(f) = 4kTG(f) \quad \text{(Power Spectral Density)}$$

- **Bipolar Junction Transistor (BJT) Noise:**
  - Base resistance thermal noise: $W_v(f) = 4kTr_{bb'}$
  - Shot noise: $W_i(f) = 2qI_{E/CO}$
  - Partition noise: $W_i(f) = 2qI_C \left(1 - \frac{|\alpha(f)|^2}{\alpha_0}\right)$
  - Flicker $(1/f)$ noise: $W_i(f) = \frac{K}{f} \cdot \frac{I}{A}$

- **MOSFET Noise & Linear System Output Noise PSD:**

$$W_{\text{out}}(f) = |H(f)|^2 W_{\text{in}}(f)$$

XII (cont.). Linear-Phase FIR Types, Design, and Digital Frequency Transformations

5 (cont.). Four Types of Linear-Phase FIR Filters

- **a. Type I:** $h(n)$ **Even-Symmetric,** N **Odd:**

$$H(e^{j\omega}) = e^{-j\frac{N-1}{2}\omega}\left\{\sum_{n=0}^{\frac{N-3}{2}} 2h(n)\cos\left[\omega\left(\frac{N-1}{2}-n\right)\right] + h\left(\frac{N-1}{2}\right)\right\}$$

  Constraint: No fixed zeros at $\omega = 0$ or π; suitable for all filter types (LP, HP, BP, BS).

- **b. Type II:** $h(n)$ **Even-Symmetric,** N **Even:** $H(e^{j\omega})\big|_{\omega=\pi} = 0$

$$H(e^{j\omega}) = e^{-j\frac{N-1}{2}\omega}\left\{\sum_{n=0}^{\frac{N}{2}-1} 2h(n)\cos\left[\omega\left(\frac{N-1}{2}-n\right)\right]\right\}$$

  Constraint: Always has a zero at $\omega = \pi$ ($z = -1$); cannot be used for highpass or bandstop filters.

- **c. Type III:** $h(n)$ **Odd-Symmetric (Anti-Symmetric),** N **Odd:** $H(e^{j\omega})\big|_{\omega=0,\pi} = 0$

$$H(e^{j\omega}) = e^{-j\frac{N-1}{2}\omega}\left\{j\sum_{n=0}^{\frac{N-3}{2}} 2h(n)\sin\left[\omega\left(\frac{N-1}{2}-n\right)\right]\right\}$$

  Constraint: Fixed zeros at both $\omega = 0$ and $\omega = \pi$; used for differentiators, Hilbert transformers, and bandpass filters.

- **d. Type IV:** $h(n)$ **Odd-Symmetric (Anti-Symmetric),** N **Even:** $H(e^{j\omega})\big|_{\omega=0} = 0$

$$H(e^{j\omega}) = e^{-j\frac{N-1}{2}\omega}\left\{j\sum_{n=0}^{\frac{N}{2}-1} 2h(n)\sin\left[\omega\left(\frac{N-1}{2}-n\right)\right]\right\}$$

  Constraint: Fixed zero at $\omega = 0$ ($z = 1$); used for differentiators, Hilbert transformers, and highpass filters.

- **2) Cascade Structure of Linear-Phase FIR Filters:** Zeros occur in mirror-image conjugate reciprocal quadruplets $(z_k, z_k^*, 1/z_k, 1/z_k^*)$ with respect to the unit circle.

6. FIR Filter Design Methods

- ① **Window Design Method (**$h(n) = h_d(n)\cdot w(n)$**):**
  - Mainlobe width $\uparrow \Longleftrightarrow$ Peak sidelobe level $\downarrow \Longleftrightarrow$ Transition band width $\uparrow \Longleftrightarrow$ Minimum stopband attenuation $\uparrow$ (stopband ripple $\downarrow$).

- ② **Frequency Sampling Design Method:**

$$H(e^{j\omega}) = \frac{1}{N}e^{-j\omega\frac{N-1}{2}}\sum_{k=0}^{N-1} H(k)e^{-j\frac{k\pi}{N}(N-1)}\frac{\sin\left[\frac{N}{2}\left(\omega-\frac{2\pi}{N}k\right)\right]}{\sin\left[\frac{1}{2}\left(\omega-\frac{2\pi}{N}k\right)\right]}$$

  Matches the desired response exactly at the frequency sampling points $\omega_k = \frac{2\pi}{N}k$. At abrupt transitions between passband and stopband, Gibbs ripples are large; typically

one inserts several unconstrained transition samples between passband and stopband to optimize stopband attenuation.

7. Digital Filter Frequency Transformations (Constantinides All-Pass Mapping $Z \leftarrow G(z)$)

Mapping requirements from prototype Z-plane to target z-plane:

- ① Unit circle of Z maps to unit circle of $G(z)$: $|G(e^{j\omega})| = 1$.
- ② Interior/boundary/exterior preserved: $|z| \gtreqless 1 \implies |G(z)| \gtreqless 1$ (Wait: for $Z = g(z)$, $|z| < 1 \implies |G(z)| < 1$ so causal stable maps to causal stable).
- ③ Rational $H(Z) \implies$ Rational $H(G(z))$.

Therefore, $G(z)$ must be an all-pass function of order N (where N equals the number of passbands of the target filter as z traverses the unit circle once):

$$G(z) = \pm \frac{D(z)}{z^N D(z^{-1})}$$

Target Filter	Mapping $Z \leftarrow G(z)$	DC ($z = 1$) Maps To	Cutoff $e^{j\theta_p}$ Maps To	Cutoff $e^{-j\theta_p}$ Maps To
Prototype	z	$z = 1$	$e^{j\theta_p}$	$e^{-j\theta_p}$
Lowpass (LP)	$\frac{z - \alpha}{1 - \alpha z}$	1	$e^{j\omega_p}$	$e^{-j\omega_p}$
Highpass (HP)	$-\frac{z + \alpha}{1 + \alpha z}$	-1	$e^{j\omega_p}$	$e^{-j\omega_p}$
Bandpass (BP)	$-\frac{z^2 + \alpha_1 z + \alpha_2}{1 + \alpha_1 z + \alpha_2 z^2}$	$e^{\pm j\omega_0}$	$e^{j\omega_2}$	$e^{j\omega_1}$
Bandstop (BS)	$\frac{z^2 + \alpha_1 z + \alpha_2}{1 + \alpha_1 z + \alpha_2 z^2}$	± 1	$e^{j\omega_1}$	$e^{j\omega_2}$

XIII. State-Space Analysis of Continuous and Discrete Systems

1. Formulation of Continuous-Time State-Space Equations

Choose integrator outputs as state variables $\boldsymbol{\lambda}(t)$ (input $\boldsymbol{e}(t)$, output $\boldsymbol{r}(t)$):

$$\begin{cases} \dfrac{d}{dt}\boldsymbol{\lambda}(t) = \boldsymbol{A}\boldsymbol{\lambda}(t) + \boldsymbol{B}\boldsymbol{e}(t) \\ \boldsymbol{r}(t) = \boldsymbol{C}\boldsymbol{\lambda}(t) + \boldsymbol{D}\boldsymbol{e}(t) \end{cases}$$

2. Solution of Continuous-Time State-Space Equations

① Time-Domain (t-Domain) Solution:

From the state equation, $\frac{d}{dt}[e^{-\boldsymbol{A}t}\boldsymbol{\lambda}(t)] = e^{-\boldsymbol{A}t}\boldsymbol{B}\boldsymbol{e}(t)$. Integrating from 0_- to t yields:

$$\boldsymbol{\lambda}(t) = e^{\boldsymbol{A}t}\boldsymbol{\lambda}(0_-) + \int_{0_-}^{t} e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{e}(\tau)\,d\tau = e^{\boldsymbol{A}t}\boldsymbol{\lambda}(0_-) + e^{\boldsymbol{A}t}\boldsymbol{B} * \boldsymbol{e}(t)$$

$$\boldsymbol{r}(t) = \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{\lambda}(0_-) + \int_{0_-}^{t} \boldsymbol{C}e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{e}(\tau)\,d\tau + \boldsymbol{D}\boldsymbol{e}(t) = \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{\lambda}(0_-) + [\boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{B} + \boldsymbol{D}\delta(t)] * \boldsymbol{e}(t)$$

② Laplace-Domain (s-Domain) Solution:

Taking the unilateral Laplace transform of the state equation gives
$s\boldsymbol{\Lambda}(s) - \boldsymbol{\lambda}(0_-) = \boldsymbol{A}\boldsymbol{\Lambda}(s) + \boldsymbol{B}\boldsymbol{E}(s)$, so:

$$\boldsymbol{\Lambda}(s) = (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{\lambda}(0_-) + (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{E}(s)$$

$$\boldsymbol{\lambda}(t) = \mathcal{L}^{-1}[(s\boldsymbol{I} - \boldsymbol{A})^{-1}]\boldsymbol{\lambda}(0_-) + \mathcal{L}^{-1}[(s\boldsymbol{I} - \boldsymbol{A})^{-1}]\boldsymbol{B} * \boldsymbol{e}(t)$$

$$\boldsymbol{r}(t) = \boldsymbol{C}\mathcal{L}^{-1}[(s\boldsymbol{I} - \boldsymbol{A})^{-1}]\boldsymbol{\lambda}(0_-) + \left\{\boldsymbol{C}\mathcal{L}^{-1}[(s\boldsymbol{I} - \boldsymbol{A})^{-1}]\boldsymbol{B} + \boldsymbol{D}\delta(t)\right\} * \boldsymbol{e}(t)$$

Under zero-state conditions ($\boldsymbol{\lambda}(0_-) = \boldsymbol{0}$), $\boldsymbol{\Lambda}(s) = (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{E}(s)$ and $\boldsymbol{R}(s) = \boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{E}(s) + \boldsymbol{D}\boldsymbol{E}(s)$, yielding the transfer function matrix and impulse response:

$$\boldsymbol{H}(s) = \boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D} = \boldsymbol{C}\frac{\operatorname{adj}(s\boldsymbol{I} - \boldsymbol{A})}{|s\boldsymbol{I} - \boldsymbol{A}|}\boldsymbol{B} + \boldsymbol{D}, \quad \boldsymbol{h}(t) = \boldsymbol{C}e^{\boldsymbol{A}t}\boldsymbol{B}u(t) + \boldsymbol{D}\delta(t)$$

- **State Transition Matrix:** $e^{\boldsymbol{A}t} = \mathcal{L}^{-1}[(s\boldsymbol{I} - \boldsymbol{A})^{-1}]$
- **Resolvent (Characteristic) Matrix:** $(s\boldsymbol{I} - \boldsymbol{A})^{-1}$

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3. Formulation of Discrete-Time State-Space Equations

Choose delay element outputs as state variables $\boldsymbol{\lambda}(n)$ (input $\boldsymbol{x}(n)$, output $\boldsymbol{y}(n)$):

$$\begin{cases} \boldsymbol{\lambda}(n+1) = \boldsymbol{A}\boldsymbol{\lambda}(n) + \boldsymbol{B}\boldsymbol{x}(n) \\ \boldsymbol{y}(n) = \boldsymbol{C}\boldsymbol{\lambda}(n) + \boldsymbol{D}\boldsymbol{x}(n) \end{cases}$$

4. Solution of Discrete-Time State-Space Equations

① Discrete Time-Domain (n-Domain) Solution:

From the state equation, $\boldsymbol{\lambda}(n)=\boldsymbol{K}\boldsymbol{A}^n u(n)+\boldsymbol{x}(n)*\boldsymbol{h}_0(n)$, solving for $\boldsymbol{K}=\boldsymbol{\lambda}(0)$ and $\boldsymbol{h}_0(n)=\boldsymbol{A}^{n-1}u(n-1)\boldsymbol{B}$ gives:

$$\boldsymbol{\lambda}(n)=\boldsymbol{A}^n\boldsymbol{\lambda}(0)u(n)+\boldsymbol{A}^{n-1}\boldsymbol{B}u(n-1)*\boldsymbol{x}(n)u(n)$$

$$\boldsymbol{y}(n)=\boldsymbol{C}\boldsymbol{A}^n\boldsymbol{\lambda}(0)u(n)+\boldsymbol{C}\boldsymbol{A}^{n-1}\boldsymbol{B}u(n-1)*\boldsymbol{x}(n)u(n)+\boldsymbol{D}\boldsymbol{x}(n)u(n)$$

② z-Domain Solution:

Taking the unilateral z-transform gives $z\boldsymbol{\Lambda}(z)-z\boldsymbol{\lambda}(0)=\boldsymbol{A}\boldsymbol{\Lambda}(z)+\boldsymbol{B}\boldsymbol{X}(z)$, so:

$$\boldsymbol{\Lambda}(z)=(z\boldsymbol{I}-\boldsymbol{A})^{-1}z\boldsymbol{\lambda}(0)+(z\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{X}(z)$$

$$\boldsymbol{\lambda}(n)=\mathcal{Z}^{-1}[(z\boldsymbol{I}-\boldsymbol{A})^{-1}z]\boldsymbol{\lambda}(0)+\mathcal{Z}^{-1}[(z\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{B}]*\boldsymbol{x}(n)$$

$$\boldsymbol{y}(n)=\boldsymbol{C}\mathcal{Z}^{-1}[(z\boldsymbol{I}-\boldsymbol{A})^{-1}z]\boldsymbol{\lambda}(0)+\left\{\boldsymbol{C}\mathcal{Z}^{-1}[(z\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{B}]+\boldsymbol{D}\delta(n)\right\}*\boldsymbol{x}(n)$$

Under zero-state conditions ($\boldsymbol{\lambda}(0)=\boldsymbol{0}$), $\boldsymbol{\Lambda}(z)=(z\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{X}(z)$ and $\boldsymbol{Y}(z)=\boldsymbol{C}(z\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{X}(z)+\boldsymbol{D}\boldsymbol{X}(z)$, yielding:

$$\boldsymbol{H}(z)=\boldsymbol{C}(z\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{B}+\boldsymbol{D}=\boldsymbol{C}\frac{\operatorname{adj}(z\boldsymbol{I}-\boldsymbol{A})}{|z\boldsymbol{I}-\boldsymbol{A}|}\boldsymbol{B}+\boldsymbol{D},\quad \boldsymbol{h}(n)=\boldsymbol{C}\boldsymbol{A}^{n-1}\boldsymbol{B}u(n-1)+\boldsymbol{D}\delta(n)$$

- **State Transition Matrix:** $\boldsymbol{A}^n=\mathcal{Z}^{-1}[(z\boldsymbol{I}-\boldsymbol{A})^{-1}z]$, and $\boldsymbol{A}^{n-1}u(n-1)=\mathcal{Z}^{-1}[(z\boldsymbol{I}-\boldsymbol{A})^{-1}]$.
- **Resolvent Matrix:** $(z\boldsymbol{I}-\boldsymbol{A})^{-1}z$.

Appendix A19: Linear Similarity Transformations of State Variables, Controllability, and Observability

1. Similarity Transformation & Transfer Function Invariance

Let $\boldsymbol{\gamma} = \boldsymbol{P\lambda}$ be a non-singular linear transformation of the state vector ($\det \boldsymbol{P} \neq 0$). The transformed state-space matrices are:

$$\boldsymbol{A}' = \boldsymbol{PAP}^{-1}, \quad \boldsymbol{B}' = \boldsymbol{PB}, \quad \boldsymbol{C}' = \boldsymbol{CP}^{-1}, \quad \boldsymbol{D}' = \boldsymbol{D}$$

The transfer function matrix is invariant under similarity transformations:

$$\boldsymbol{H}'(s/z) = \boldsymbol{C}'(s\boldsymbol{I} - \boldsymbol{A}')^{-1}\boldsymbol{B}' + \boldsymbol{D}' = \boldsymbol{H}(s/z)$$

2. Jordan Canonical Form & Kalman Controllability / Observability Criteria

Using the modal matrix $\boldsymbol{P}$ formed by the eigenvectors (and generalized eigenvectors) of $\boldsymbol{A}$ from its characteristic equation, transform $\boldsymbol{A}$ into its **Jordan Canonical Form** $\boldsymbol{A}' = \boldsymbol{J}$:

- **Controllability Criterion:**
  - **Gilbert / Jordan Form Criterion:** For each distinct eigenvalue's Jordan blocks, the rows of $\boldsymbol{B}' = \boldsymbol{PB}$ corresponding to the *last row* of each Jordan sub-block contain no all-zero row (and for distinct blocks sharing the same eigenvalue, those last rows are linearly independent).
  - **Kalman Rank Criterion:** Equivalent to the controllability matrix $\boldsymbol{M}$ having full row rank N:

$$\operatorname{rank} \boldsymbol{M} = \operatorname{rank} \begin{bmatrix} \boldsymbol{B} & \boldsymbol{AB} & \cdots & \boldsymbol{A}^{N-1}\boldsymbol{B} \end{bmatrix} = N \iff \text{System is State Controllable.}$$

- **Observability Criterion:**
  - **Gilbert / Jordan Form Criterion:** For each Jordan sub-block, the columns of $\boldsymbol{C}' = \boldsymbol{CP}^{-1}$ corresponding to the *first column* of each Jordan sub-block contain no all-zero column (and for blocks sharing the same eigenvalue, those first columns are linearly independent).
  - **Kalman Rank Criterion:** Equivalent to the observability matrix $\boldsymbol{N}$ having full column rank N:

$$\operatorname{rank} \boldsymbol{N} = \operatorname{rank} \begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{CA} \\ \vdots \\ \boldsymbol{CA}^{N-1} \end{bmatrix} = N \iff \text{System is State Observable.}$$